

EEBus UC Technical Specification

Monitoring of Power Consumption

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EEBus Initiative e.V.

Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

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1 Scope of the document

This document describes the Use Case "Monitoring of Power Consumption" (short-name: MPC). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or an electric vehicle).

GCP

Abbreviation for "Grid Connection Point"

MPC

Monitoring of Power Consumption (short-name of this Use Case)

RMS (rms)

Abbreviation for "root mean square" (used for electricity measurement)

Scenario

Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or optional.

Specialization

Reusable data collection for a specific functionality.

SPINE

Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative e.V.

1.3 Requirements

1.3.1 Requirements wording

The following keywords are used:

- SHALL
- SHALL NOT
- SHOULD
- SHOULD NOT
- MAY

Note: They apply only if written in capital letters.

For the meaning of the keywords, please refer to [RFC2119].

1.3.2 Mapping of High-Level requirements

Within the High-Level Use Case description, the following abbreviation is used:

[MPC-xyz]

e.g.: [MPC-007]

The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique number xyz. These requirements are referenced throughout the technical solution to show how each High-Level requirement is realized in the technical part.

2 High-Level description

2.1 Introduction

Within an overall energy management concept, it is important for the customer energy manager (CEM) to know about the electrical power consumption or production of connected devices. This holds valid not only for complex energy consumers or producers that are manageable through incentive tables or power sequences, but also for simple devices that may be switched on and off or are even un-configurable but need to be considered as energy consumers within the house or premises.

The more complex energy consumers that offer flexibility via power sequences or accept incentives to adapt their power consumption according to the recommendations of an energy manager, often predict their power consumption but may deviate therefrom. To track the real power consumption, this Use Case may be used.

Energy producers can be monitored with this Use Case as well. Devices that are able to consume and produce energy are supported, too.

Additionally, the consumed and produced energy, the current consumption, the voltage and the frequency may be offered by the Actor Monitored Unit.

Monitored Units may be connected to more than one phase of the grid connection point of the house or premises. In this case, the power measurands can be provided for the individual phases, but a device is not obliged to offer these phase-specific values (the total-value is independent from that choice and SHALL always be provided). The current and voltage values are always phase-specific and are only provided if the Monitored Unit is aware of its individual connected phases.

A CEM SHALL NOT act as Actor Monitored Unit within this Use Case.

Following the "load convention" (i.e. "passive sign convention"), measured electrical current, active power and exchanged energy are expressed with positive values in case of energy consumption whereas negative values are used in case of energy production. Voltages are measured independent of the energy direction. See section 2.5.1 for details.

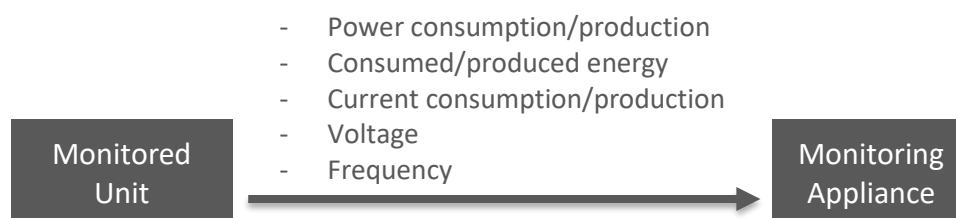


Figure 1: High-Level Use Case functionality overview

Note: This Use Case does not limit the number of logical Use Case instances allowed on the device. Please check (national) regulatory requirements on the number of logical Use Case instances allowed on a device.

206 *Added value:* The Monitoring Appliance (e.g. a CEM) knows about the power consumption or
207 production of the Monitored Unit and is able to integrate that knowledge into its overall energy
208 management.

209

210 **2.2 Actors**

211 **2.2.1 Monitoring Appliance**

212 The Actor Monitoring Appliance (e.g. a CEM) can manage the home's appliances to ensure energy
213 optimized operation of the appliances or visualize the information. In this Use Case, the Monitoring
214 Appliance retrieves power consumption or production data provided by a Monitored Unit.

215

216 **2.2.2 Monitored Unit**

217 The Actor Monitored Unit represents a device consuming or producing energy and performing some
218 functionality that is not described within this Use Case. The energy consumption or production may
219 be on up to three phases. Any energy consuming or producing device can offer this Use Case.

220

2.3 Scenarios

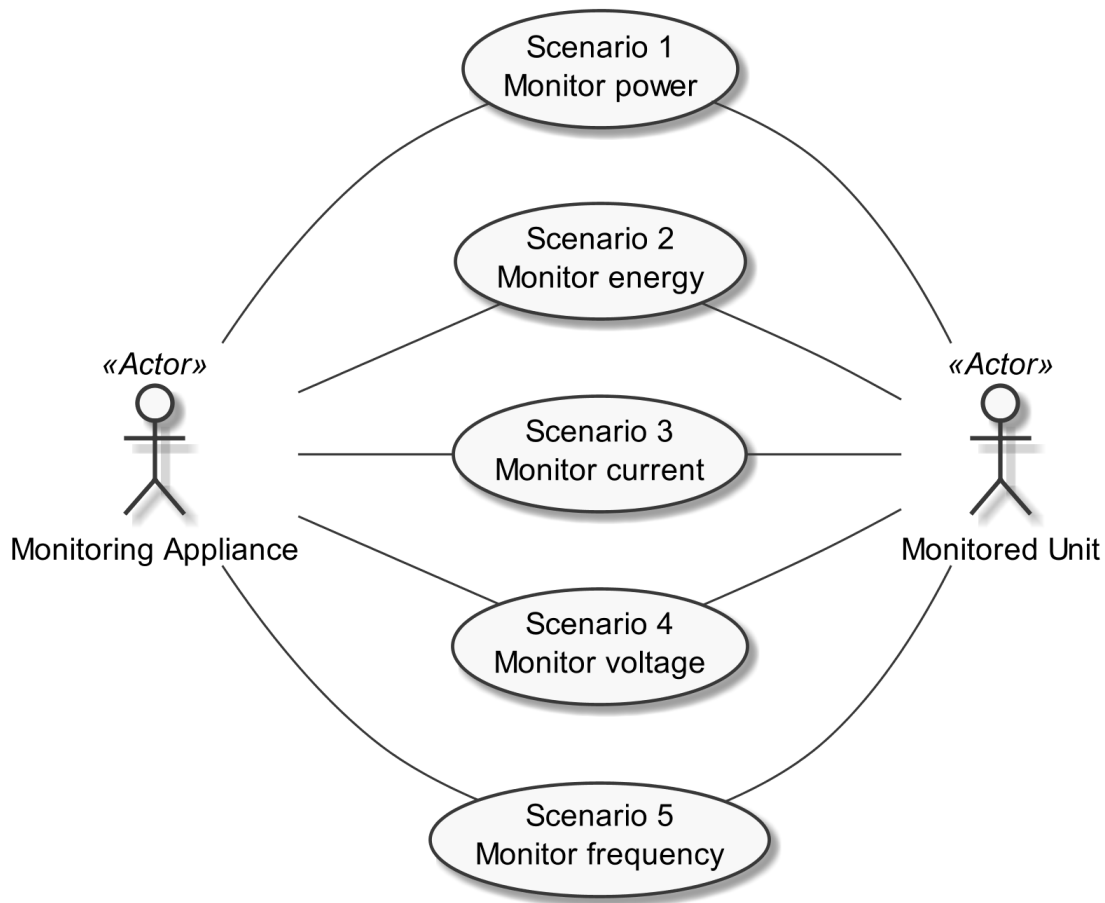


Figure 2: Scenario overview

Scenario number	Scenario name	Monitoring Appliance	Monitored Unit
1	Monitor power	M	M
2	Monitor energy	O	O
3	Monitor current	R	R ^{*1}
4	Monitor voltage	O	O
5	Monitor frequency	O	O

Table 1: Scenario implementation requirements for Actors

^{*1}: The current values MAY only be provided if the Actor Monitored Unit is aware of its connected phases (this may be only one phase or two phases, too). If the Actor Monitored Unit is aware of its connected phases, the phase specific current consumption SHOULD be provided.

2.3.1 Scenario 1 - Monitor power

2.3.1.1 Description

The momentary active power consumption or production is provided by this Scenario. Monitored Units that are connected to more than one phase may additionally provide the phase-specific values. Only AC power is considered in this Use Case.

Data point name	Data point description	Support indication	High-Level requirement
Total Active Power	The total active power of the Monitored Unit.	M	[MPC-011]
Phase-Specific Active Power	The phase-specific active power: - Active power on phase A ○ [MPC-012/1] - Active power on phase B ○ [MPC-012/2] - Active power on phase C ○ [MPC-012/3]	O* ¹	[MPC-012]

Table 2: Scenario 1 - Monitor power consumption - Data point list

*¹: The phase-specific active power values MAY only be provided if the Actor Monitored Unit is aware of its connected phases (this may be only one phase or two phases, too).

2.3.1.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the actual active power value(s) of the Actor Monitored Unit.

Pre-condition:

The Actor Monitoring Appliance does not know the actual active power value(s) of the Actor Monitored Unit.

Post-condition:

The Actor Monitoring Appliance knows the actual active power value(s) of the Actor Monitored Unit.

2.3.2 Scenario 2 - Monitor energy

2.3.2.1 Description

The total energy consumed by the Actor Monitored Unit can be provided with this Scenario.

Data point name	Data point description	Support indication	High-Level requirement
Total Consumed Energy	The total energy consumed by the Monitored Unit since installation or last reset.	M* ¹	[MPC-021]
Total Produced Energy	The total energy produced by the Monitored Unit since installation or last reset.	M* ²	[MPC-022]

Table 3: Scenario 2 - Monitor consumed energy - Data point list

*¹: SHALL only be provided if the Monitored Unit is capable of consuming energy.

*²: SHALL only be provided if the Monitored Unit is capable of producing energy.

2.3.2.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the total consumed or produced energy of the Actor Monitored Unit.

Pre-condition:

The Actor Monitoring Appliance does not know the total consumed or produced energy of the Actor Monitored Unit.

Post-condition:

The Actor Monitoring Appliance knows the total consumed or produced energy of the Actor Monitored Unit.

2.3.3 Scenario 3 - Monitor current

2.3.3.1 Description

The momentary current consumption or production is provided by this Scenario. Only Monitored Units that are connected to more than one phase or are aware of the phase(s) they are connected to, may provide the phase-specific value(s) ([MPC-031]) of this Scenario. Only the active current is considered in this Use Case ([MPC-030/1]). Only rms values are considered ([MPC-030/2]).

Data point name	Data point description	Support indication	High-Level requirement
Phase-Specific AC Current	The phase-specific active AC current: - Current on phase A ○ [MPC-031/1] - Current on phase B ○ [MPC-031/2] - Current on phase C ○ [MPC-031/3]	M	[MPC-031]

Table 4: Scenario 3 - Monitor current - Data point list

2.3.3.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the actual AC current value(s) of the Actor Monitored Unit.

Pre-condition:

The Actor Monitoring Appliance does not know the actual AC current value(s) of the Actor Monitored Unit.

Post-condition:

The Actor Monitoring Appliance knows the actual AC current value(s) of the Actor Monitored Unit.

2.3.4 Scenario 4 - Monitor voltage**2.3.4.1 Description**

The actual phase-specific AC voltage values of the Actor Monitored Unit are provided by this Scenario. Depending on the number of connected phases*, the Monitored Unit provides a different amount of individual values.

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Phase-Specific AC Voltage	Voltage between phase A and neutral [MPC-041/1].	M* ¹	[MPC-041]
	Voltage between phase B and neutral [MPC-041/2].	M* ¹	
	Voltage between phase C and neutral [MPC-041/3].	M* ¹	
	Voltage between phase A and phase B [MPC-041/4].	O* ¹	
	Voltage between phase B and phase C [MPC-041/5].	O* ¹	
	Voltage between phase C and phase A [MPC-041/6].	O* ¹	

Table 5: Scenario 4 - Monitor voltage - Data point list

*¹: Only values related to the connected phases SHALL be delivered.

2.3.4.2 Conditions**Triggering Event:**

The Actor Monitoring Appliance is interested in the phase-specific AC voltage values of the Actor Monitored Unit.

Pre-condition:

The Actor Monitoring Appliance does not know the phase-specific AC voltage values of the Actor Monitored Unit.

Post-condition:

The Actor Monitoring Appliance knows the phase-specific AC voltage values of the Actor Monitored Unit.

2.3.5 Scenario 5 - Monitor frequency**2.3.5.1 Description**

The frequency at the AC connection of the Monitored Unit is provided by this Scenario.

Data point name	Data point description	Support indication	High-Level requirement
AC Frequency	Frequency, the Monitored Unit measures at its AC connection.	M	[MPC-051]

Table 6: Scenario 3 - Monitor frequency - Data point list

2.3.5.2 Conditions

Triggering Event:

The Actor Monitoring Appliance is interested in the AC frequency of the Actor Monitored Unit.

Pre-condition:

The Actor Monitoring Appliance does not know the AC frequency of the Actor Monitored Unit.

Post-condition:

The Actor Monitoring Appliance knows the AC frequency of the Actor Monitored Unit.

2.4 Dependencies to other Use Cases

2.4.1 "Monitoring of Inverter"

In case of an implementation of this Use Case on an Inverter, the Use Case "Monitoring of Inverter" SHALL be considered, especially the rules regarding the resource hierarchy of the Inverter SHALL be followed.

2.5 Further information and rules

2.5.1 Sign conventions

Throughout the whole Use Case the "load convention" (i.e. "passive sign convention") is applied for measurement values ([MPC-001]). This means measured electrical current, active power and exchanged energy are expressed with positive values in case of energy consumption whereas negative values are used in case of energy production. This holds valid for the following values of this Use Case:

- Total Active Power ([MPC-011])
- Phase-Specific Active Power ([MPC-012])
- Total Consumed Energy ([MPC-021])
- Total Produced Energy ([MPC-022])
- Phase-Specific AC Current ([MPC-031])

The Monitored Unit's voltages ([MPC-041]) are measured independent of the energy direction ([MPC-002]).

2.5.2 Value state rules

Each Monitored Unit's measurement value has a state that defines whether the value is correct or is to be ignored by the Monitoring Appliance ([MPC-003]):

- normal: value correct
- out of range: value is out of range and SHALL be ignored by the Monitoring Appliance
- error: value is erroneous and SHALL be ignored by the Monitoring Appliance.

2.6 Assumptions and Prerequisites

None.

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.2.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.2.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "monitoringOfPowerConsumption". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
--------------	---------	------------------------------

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 7: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

420 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 421 - M, O
- 422 - M \W, O \W

423 The description (rules) of the Element details in such a case the presence requirement. A typical
424 example is a list-based Element where the "first" (or "last") Element has a different presence
425 requirement than the other list Elements.

426

427 **3.1.3.3 Presence indications for "Content of Function..." tables**

428 This section only defines rules for the server side.

429 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
430 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
431 Element, because the Element is set only by the client.

432 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 433 - "R" means that the data SHOULD be provided by the server.
- 434 - "O" means that the data MAY be provided by the server.
- 435 - "F" means that the data SHALL NOT be provided by the server.

436 The following applies for writeable data that is exchanged in a "write" operation:

- 437 - "R" means that the data SHOULD be supported. In other words: If the client writes the
438 Element, the server SHOULD accept those messages and the contained Elements.
- 439 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
440 the server MAY accept those messages and the contained Elements.

441 The following applies for Elements that are not listed in the Actor section:

- 442 - In case of a received "read" request: The according Element MAY be set in the reply.
- 443 - In case of a received "write" operation: The server MAY ignore the Element.
- 444 - In case of a "notify" operation to be created: The server MAY set the Element.

445 Note: The server will only accept write operations if the result fulfils the server Function
446 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
447 result in an error message.

448 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
449 only has to be supported during a certain event and hence does not need to be present at all times. If
450 the event is not active the Element may be omitted or another value may be set. In most cases a
451 High-Level requirement reference for the event is given in the rules column.

452 In some cases, an Element may have a double presence indication, separated by a colon. Examples:

- 453 - M, O
- 454 - M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
No cardinality indication.
2. X (a..b)
This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..
This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 8: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3

times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#{1..}>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

552 Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g.
 553 In this case, the uniqueness rule applies for "<p5>" likewise.

554 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

555

556 3.1.3.6.3 Placeholder expressions with cardinalities

557 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 558 purpose, which make use of cardinality expressions. The general notation is as follows:

559 1. <xM#(a..b)>

560 This is equivalent to this explicit definition:

561 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

562 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 563 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 564 respective Identifier.

565 Additionally, the following notations may occur:

566 2. <xM#(a..)>

567 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

568 3. <xM#(..b)>

569 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

570 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 571 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 572 "b" distinct values.

573 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*(a..b)> etc.

574 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 575 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 576 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 577 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 578 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 579 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

580

581 3.1.3.6.4 References to placeholders and relations

582 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 583 to a particular placeholder:

584 1. e(-><xM>)

585 2. e(-><xM#S>)

586 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using this expression:

"a"(-><m2#1>)

"b"(-><m2#2>)

"c"(-><m2#3>)

This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to be used with <m2#2> and "c" with <m2#3>.

Similarly, a placeholder "yN" can refer to a particular placeholder:

3. <yN->xM>

4. <yN->xM#S>

5. <yN#T->xM>

6. <yN#T->xM#S>

where "T" is a sub-number of "yN".

It is also feasible to associate placeholders with cardinalities:

7. <yN#(a..b)->xM#(a..b)>

denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

In some cases, there is a need to express that multiple list items with similar values are feasible or required, but only particular combinations of these different data are permitted. The following example shows that several "fData" list items with different "phase" value are required, but that these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase" value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a table:

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

Table 9: Content of an example Specialization

Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or {<z3#1>, <z3#4>, <z3#5>}.

3.1.3.7 Rules for content of "Value" column

For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element to one or more particular values. This means that Elements with values deviating from the restriction (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered as if the Element is not set. If more than one particular value is permitted for an Element, the values are in a single line each.

If a presence indication is set for the value (in an additional column before the value), the following rules SHALL be applied:

- "M" means that the value SHALL be supported. This means the value needs to be set at a certain point in time (depending on the value rules) or for a certain Element within a list entry.
- "R" means that the value SHOULD be supported.
- "O" means that the value MAY be supported.

If all possible values of a given mandatory Element are optional or recommended and this Element is used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values of a given optional or recommended Element are optional or recommended, this Element MAY contain also other values, but then this Element SHALL NOT be considered as part of the respective Scenario.

M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

If no presence indication is set for the value, the following rules SHALL be applied:

- In case of Elements where the server may set or change an Element on its own (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support at least one of the listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support all listed values.
- In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - o within the tables in the "Server data - Resources" sections:
 - the server SHALL support all listed values.
 - o within the tables in the "Client data - Specializations" sections:
 - the client SHALL support at least one of the listed values.

Depending on the Element, different values may be used during runtime. If this is the case, those rules are described within the value rules.

651 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 652 MAY deviate from this, e.g. "(1024)".

653

654 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 655 **Specialization..." tables**

656 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 657 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 658 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 659 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 660 background colour. This row contains the following parts:

661 <Feature Type>. <Function>.[<list entry instance name>.]

662 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 663 could be:

664 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 665 deviceConfigurationKeyValueDescriptionData.

666 In the following rows, only the names of the Elements are stated, without the prefix described above.

667

668 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 669 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 670 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 671 "Function "measurementListData"). These sections contain one table with all Elements needed for
 672 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 673 colour. This row contains the following parts:

674 <Feature Type>. <Function>.[<list entry instance name>.]

675 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 676 could be:

677 Measurement. measurementDescriptionListData. measurementDescriptionData.

678 In the following rows, only the names of the Elements are stated, without the prefix described above.

679

680 For both kinds of tables, the following applies:

681 - Parent Elements are marked with a dot at the end of the name:

682 <parent Element>.

683 E.g.:

684 value.

685 - If there are sub-Elements, they are described in own rows with the name of the parent
 686 Element as prefix, separated by a dot and a blank space:

687 <parent Element>. <sub-Element>
 688 E.g.:
 689 value. number

690

691 3.1.4 Rules for "Feature Types and Functions..." tables

692 3.1.4.1 Presence indications for "Feature Types and Functions..." tables

693 The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

694 Table 10: Presence indication of Feature Types and Functions support

695 If at least one Function of a Feature has the presence indication "M", it is mandatory to support the
 696 Feature.

697

698 3.1.4.2 Rules for "Possible operations" column

699 Within the "Feature Types and Functions..." tables the column "Possible operations" state whether
 700 the Function is read- or writeable (as defined in the detailed discovery mechanism, see
 701 [ProtocolSpecification]).

702 If the "partial" concept (also called "restricted function exchange") SHALL be supported, the
 703 following notation is used (separated for read and write access):

704 read (M). partial (M)
 705 write (M). partial (M)

706 If the "partial" concept SHOULD be supported, the following notation is used:

707 read (M). partial (R)
 708 write (M). partial (R)

709 If the "partial" concept MAY be supported, the following notation is used:

710 read (M). partial (O)
 711 write (M). partial (O)

712 The server can decide whether a notification is submitted complete or partial (as described in
 713 [ProtocolSpecification]) if not defined differently within this Use Case Specification.

714

715 3.1.5 "Actor ... overview" diagram rules

716 Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of
 717 this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in
 718 Table 1. The Actor MAY have more functionality implemented than needed for this Use Case.

For the following Actor overview example, a brief description of the graphical symbols will be described.

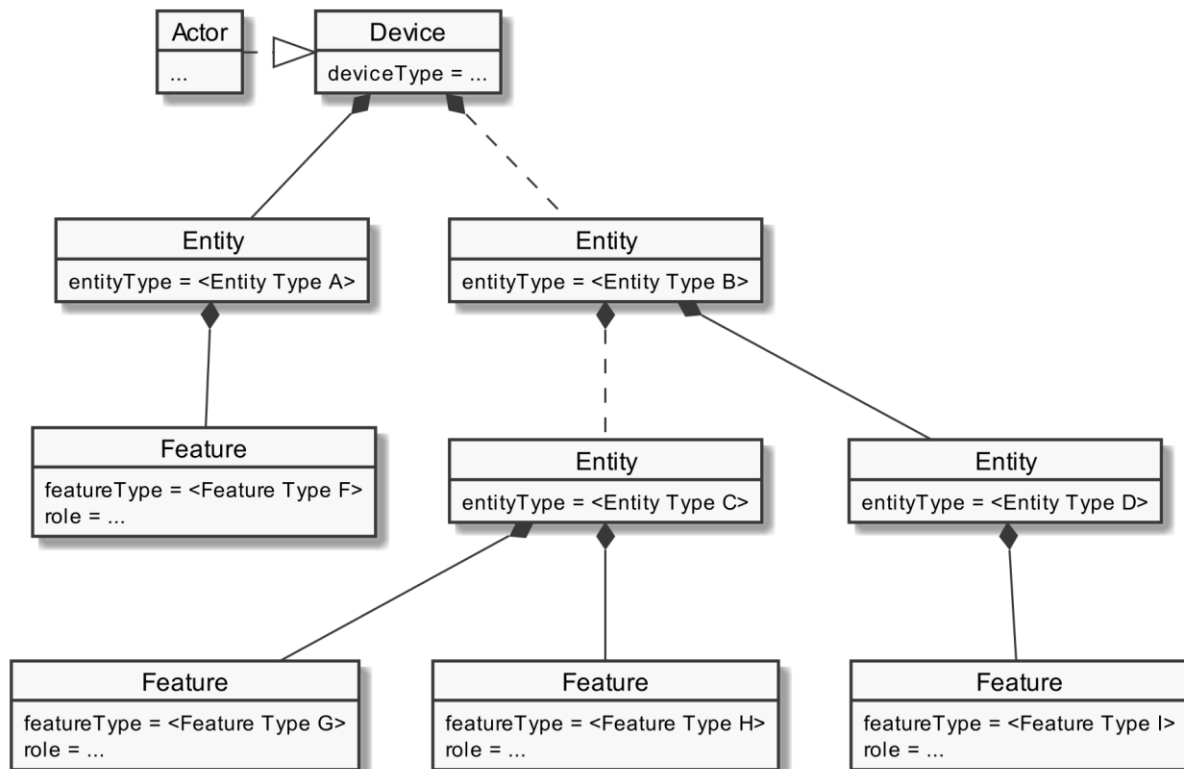


Figure 3: Actor overview example

The solid lines in the figure represent an immediate parent-childhood relation: The Entity with "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

The dashed lines in the figure express that there MAY be additional Entities between the shown Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

3.1.6 Specializations

Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a Specialization defines a dataset that may encompass multiple related Functions from one or more different Features.

As different Use Cases sometimes share similar requirements, Specializations are also important from a re-usability perspective. This approach is used to improve consistency across Use Cases and avoid multiple variances of basically the same dataset. This is especially important in the case when an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Measurement value rules]", e.g.

[Measurement value rules]:

SHALL be set if a value is available. Otherwise the whole list entry SHALL be omitted or the Element *valueState* SHALL be set to "error".

If *valueState* is set to "error", but *value* is set, the content of *value* SHALL be ignored.

If *valueState* is set to "outOfRange", but *value* is set, the content of *value* SHALL be interpreted as being out of range.

If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMax* is set and *value* is equal or bigger than *valueRangeMax*, *value* SHALL be interpreted as above *valueRangeMax*.

779 If *valueState* is set to "outOfRange", *measurementConstraintsListData.valueRangeMin* is set and
780 *value* is equal or smaller than *valueRangeMin*, *value* SHALL be interpreted as below *valueRangeMin*.

781 If set, *measurementDescriptionListData.measurementType* SHALL be set, too.

782

783 **[Value state rules]:**

784 The Element *valueState* SHALL be set if its content differs from "normal" ([MPC-003]). This means, if
785 the state of the value is "outOfRange" or "error" this SHALL be denoted in the *valueState* Element. A
786 client SHALL always consider the content of *valueState*, if set. If omitted, a value of "normal" is
787 assumed.

788

789 **[String-length rules]:**

790 The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL
791 consider the possibility that the receiver will shorten the string to 256 characters.

792

793 **[Scaled number rules]:**

794 The sub-Elements "number" and "scale" represent a value according to the following formula:
795 $\text{number} * 10^{\text{scale}}$

796

797 **[Evaluation period rules]:**

798 The "evaluationPeriod" expresses the analysis period for the according value.

799

800 **3.1.8.2 Rules regarding the usage of time-related information**

801 All time-related information within this Use Case SHALL be presented as absolute times. This means
802 relative times SHALL NOT be used, unless the data point only supports relative times (e.g. for a
803 "duration").

804 Note: This means the Actors need to have a common understanding of these absolute times. It is
805 strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated
806 Universal Time" (UTC).

807

808 **3.1.8.3 Applied sign convention**

809 For the applied sign convention, please see section 2.5.1.

810

3.2 Actors

3.2.1 Monitoring Appliance

3.2.1.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "MonitoringAppliance" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Monitoring Appliance's resource hierarchy.

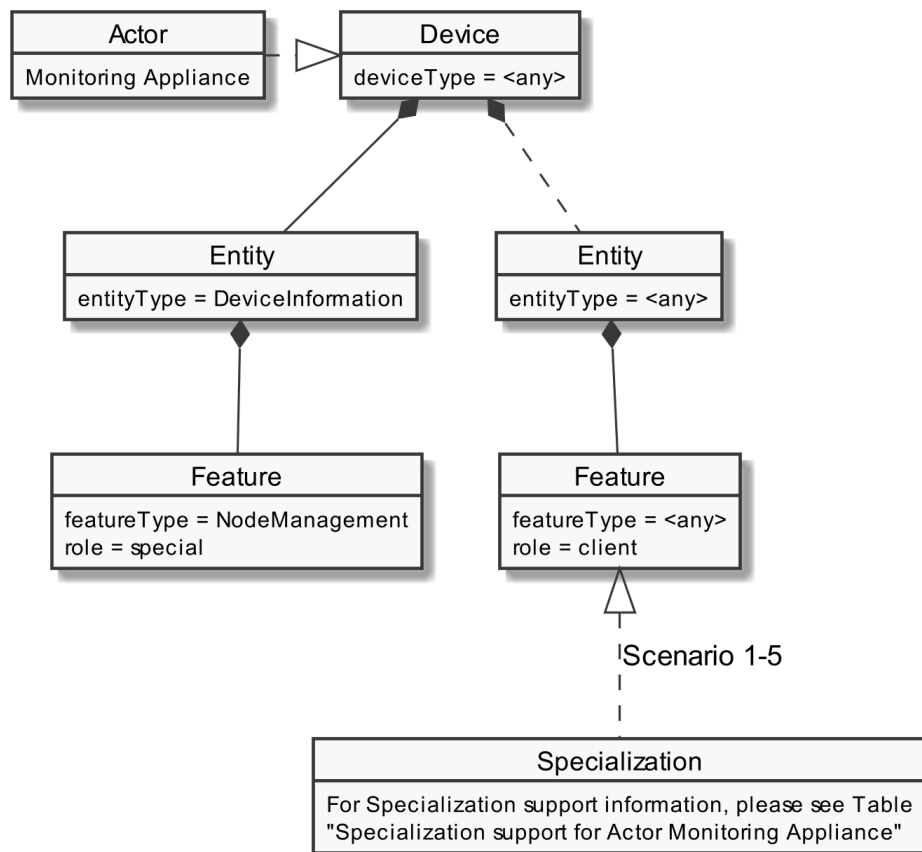


Figure 4: Actor "Monitoring Appliance" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follows behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a

corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
Measurement_AcPowerTotal	1	Table 12
Measurement_AcPowerRealPhaseSpecific	1	Table 13
Measurement_AcEnergyConsumed	2	Table 14
Measurement_AcEnergyProduced	2	Table 15
Measurement_AcCurrentPhaseSpecific	3	Table 16
Measurement_AcVoltagePhaseSpecific	4	Table 17
Measurement_AcFrequency	5	Table 18

Table 11: Specialization support for Actor Monitoring Appliance

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "Measurement"

3.2.1.3.1.1 Specialization "Measurement_AcPowerTotal"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	measurementType	"power"	
1: M	commodityType	"electricity"	
1: M	unit	"W"	
1: M	scopeType	"acPowerTotal"	
1: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: R	valueRangeMin.		[MPC-001] SHOULD be used. See [Scaled number rules].
1: M	valueRangeMin. number		SHALL be used.
1: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.

1: R	valueRangeMax.		[MPC-001] SHOULD be used. See [Scaled number rules].
1: M	valueRangeMax. number		SHALL be used.
1: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	valueStepSize.		SHOULD be used. See [Scaled number rules].
1: M	valueStepSize. number		SHALL be used.
1: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	Measurement. measurementListData. measurementData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
1: O	timestamp	<t#{1..1}->m1#1>	MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
1: M	value.		[MPC-001], [MPC-011] See [Measurement value rules]. See [Scaled number rules].
1: M	value. number		SHALL be used.
1: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: M	valueState		[MPC-003] [Value state rules]
1: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	powerSupplyType	"ac"	
1: M	positiveEnergyDirection	"consume"	
1: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	parameterId	<p1#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
1: M	measurementId	<m1#{1..1}->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
1: M	voltageType	"ac"	
1: O	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c" (->p1#1)	If the Monitored Unit is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is connected to the Monitored Unit.

1: O	acMeasuredInReferenceTo	"neutral"	
1: M	acMeasurementType	"real"	
1: O	acMeasurementVariant	"rms"	

Table 12: Content of Specialization "Measurement_AcPowerTotal" at Actor Monitoring Appliance

3.2.1.3.1.2 Specialization "Measurement_AcPowerRealPhaseSpecific"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
1: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	measurementType	"power"	
1: M	commodityType	"electricity"	
1: M	unit	"W"	
1: M	scopeType	"acPower"	
1: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
1: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
1: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMin. number		SHALL be used.
1: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMax. number		SHALL be used.
1: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	valueStepSize.		SHOULD be used. [Scaled number rules]
1: M	valueStepSize. number		SHALL be used.
1: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	Measurement. measurementListData. measurementData.		
1: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
1: O	timestamp	<t#(1..1)->m2#(1..3)>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
1: M	value.		[MPC-001], [MPC-012] [Measurement value rules]

			[Scaled number rules]
1: M	value. number		SHALL be used.
1: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: M	valueState		[MPC-003] [Value state rules]
1: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	powerSupplyType	"ac"	
1: M	positiveEnergyDirection	"consume"	
1: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	parameterId	<p2#{1..3}->ec1#1>	SHALL be set as SUB IDENTIFIER.
1: M	measurementId	<m2#{1..3}->p2#{1..3}>	SHALL be set as FOREIGN IDENTIFIER.
1: M	voltageType	"ac"	
1: M	acMeasuredPhases	"a" (->p2#1)	[MPC-012/1]
		"b" (->p2#2)	[MPC-012/2]
		"c" (->p2#3)	[MPC-012/3]
1: M	acMeasuredInReferenceTo	"neutral"	
1: M	acMeasurementType	"real"	
1: M	acMeasurementVariant	"rms"	

Table 13: Content of Specialization "Measurement_AcPowerRealPhaseSpecific" at Actor Monitoring Appliance

3.2.1.3.1.3 Specialization "Measurement_AcEnergyConsumed"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"acEnergyConsumed"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]

2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#{1..1}->m3#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MPC-001], [MPC-021] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: O	evaluationPeriod.		[Evaluation period rules]
2: M	evaluationPeriod. startTime		
2: M	evaluationPeriod. endTime		
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MPC-003] [Value state rules]
2: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	powerSupplyType	"ac"	
2: M	positiveEnergyDirection	"consume"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p3#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m3#{1..1}->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: M	acMeasurementType	"real"	

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851 3.2.1.3.1.4 Specialization "Measurement_AcEnergyProduced"

Scenario [...]: M/R/O [W][V]	Element	Value	[High-Level Mapping] Element and value rules
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"acEnergyProduced"	
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#{1..1}->m4#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MPC-001], [MPC-022] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: O	evaluationPeriod.		[Evaluation period rules]

2: M	evaluationPeriod. startTime		
2: M	evaluationPeriod. endTime		
2: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: M	valueState		[MPC-003] [Value state rules]
2: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	powerSupplyType	"ac"	
2: M	positiveEnergyDirection	"consume"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p4#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m4#{1..1}->p4#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: M	acMeasurementType	"real"	

Table 15: Content of Specialization "Measurement_AcEnergyProduced" at Actor Monitoring Appliance

3.2.1.3.1.5 Specialization "Measurement_AcCurrentPhaseSpecific"

Scenario [{...}]: M/R/O [W][V]	Element	Value	[High-Level Mapping] Element and value rules
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m5#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"current"	
3: M	commodityType	"electricity"	
3: M	unit	"A"	
3: M	scopeType	"acCurrent"	
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m5#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]

3: M	valueRangeMax. number		SHALL be used.
3: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m5#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m5#(1..3)>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MPC-001], [MPC-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: M	valueState		[MPC-003] [Value state rules]
3: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	powerSupplyType	"ac"	
3: M	positiveEnergyDirection	"consume"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p5#(1..3)->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m5#(1..3)->p5#(1..3)>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases* ²	"a" (-><p5#1>)* ¹	[MPC-031/1]
		"b" (-><p5#2>)* ¹	[MPC-031/2]
		"c" (-><p5#3>)* ¹	[MPC-031/3]
3: M	acMeasurementType	"real"	[MPC-030/1]
3: M	acMeasurementVariant	"rms"	[MPC-030/2]

Table 16: Content of Specialization "Measurement_AcCurrentPhaseSpecific" at Actor Monitoring Appliance

*¹: Each permitted value of the Element "acMeasuredPhases" SHALL NOT be used for more than one value of Element "parameterId".

*²: Only the connected phases SHALL be delivered. If only one phase is connected, only one of the values is provided by the Monitored Unit. This is not necessarily phase A, but could also be phase B or phase C.

3.2.1.3.1.6 Specialization "Measurement_AcVoltagePhaseSpecific"

Scenario [...]: M/R/O [W][\C]	Element	Value	[High-Level Mapping] Element and value rules
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"voltage"	
4: M	commodityType	"electricity"	
4: M	unit	"V"	
4: M	scopeType	"acVoltage"	
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MPC-002] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MPC-002] SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#{1..1}->m6#{1..6}>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MPC-002], [MPC-041] [Measurement value rules] [Scaled number rules]

4: M	value. number		SHALL be used.
4: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
4: R	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
4: M	valueState		[MPC-003] [Value state rules]
4: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	powerSupplyType	"ac"	
4: M	positiveEnergyDirection	"consume"	
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p6#(1..6)->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#(1..6)->p6#(1..6)>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasuredPhases	"a" (->p6#1>)	[MPC-041/1]
		"a" (->p6#4>)	[MPC-041/4]
		"b" (->p6#2>)	[MPC-041/2]
		"b" (->p6#5>)	[MPC-041/5]
		"c" (->p6#3>)	[MPC-041/3]
		"c" (->p6#6>)	[MPC-041/6]
4: M	acMeasuredInReferenceTo	"a" (->p6#6>)	[MPC-041/6]
		"b" (->p6#4>)	[MPC-041/4]
		"c" (->p6#5>)	[MPC-041/5]
		"neutral" (->p6#1>)	[MPC-041/1]
		"neutral" (->p6#2>)	[MPC-041/2]
		"neutral" (->p6#3>)	[MPC-041/3]
4: M	acMeasurementType	"apparent"	
4: M	acMeasurementVariant	"rms"	

Table 17: Content of Specialization "Measurement_AcVoltagePhaseSpecific" at Actor Monitoring Appliance

Note: The Specialization permits up to six phase measurements: Measurement of phase "a" to "neutral", phase "a" to phase "b", phase "b" to "neutral", etc.

3.2.1.3.1.7 Specialization "Measurement_AcFrequency"

Scenario {...}]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		

5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"frequency"	
5: M	commodityType	"electricity"	
5: M	unit	"Hz"	
5: M	scopeType	"acFrequency"	
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: M	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: M	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: M	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m7#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#(1..1)->m7#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MPC-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: M	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueSource	"measuredValue" "calculatedValue" "empiricalValue"	
5: M	valueState		[MPC-003] [Value state rules]
5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
5: M	powerSupplyType	"ac"	
5: M	positiveEnergyDirection	"consume"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		

5: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p7#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m7#{1..1}->p7#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"ac"	

Table 18: Content of Specialization "Measurement_AcFrequency" at Actor Monitoring Appliance

3.2.2 Monitored Unit

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "MonitoredUnit" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor Monitored Unit's resource hierarchy.

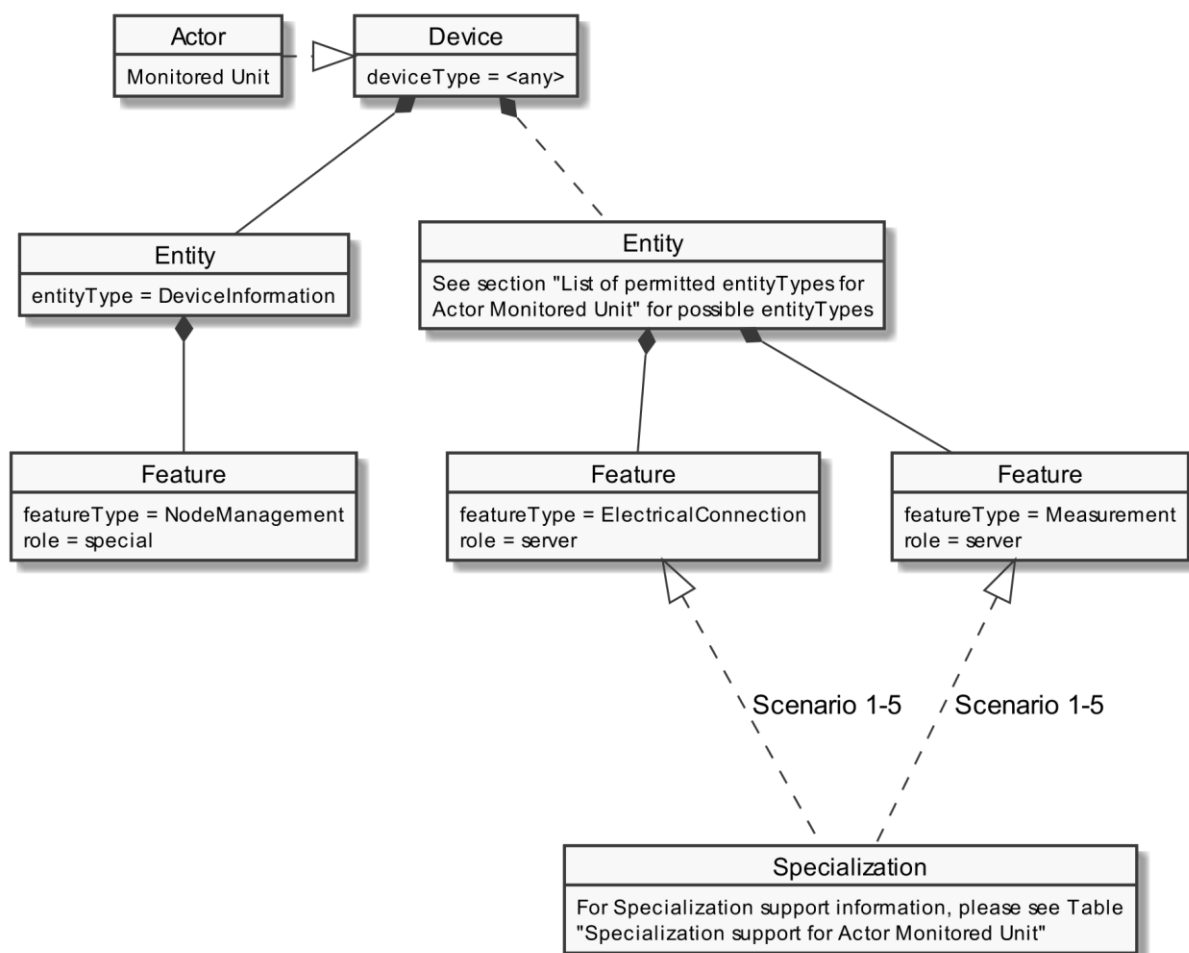


Figure 5: Actor "Monitored Unit" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind one of the entityTypes listed in section 3.2.2.1.1. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
Measurement_AcPowerTotal	1	ElectricalConnection	Table 21
			Table 22
		Measurement	Table 23
			Table 24
Measurement_AcPowerRealPhaseSpecific	1	ElectricalConnection	Table 25
			Table 21
		Measurement	Table 22
			Table 23
Measurement_AcEnergyConsumed	2	ElectricalConnection	Table 24
			Table 25
		Measurement	Table 21
			Table 22
Measurement_AcEnergyProduced	2	ElectricalConnection	Table 23
			Table 24
		Measurement	Table 25
			Table 21
Measurement_AcCurrentPhaseSpecific	3	ElectricalConnection	Table 22
			Table 21
		Measurement	Table 23
			Table 24
Measurement_AcVoltagePhaseSpecific	4	ElectricalConnection	Table 25
			Table 21
		Measurement	Table 22
			Table 23
Measurement_AcFrequency	5	ElectricalConnection	Table 24
			Table 25

		Measurement	Table 23
			Table 24
			Table 25

Table 19: Specialization support for Actor Monitored Unit

3.2.2.1.1 List of permitted entityTypes for Actor Monitored Unit

In this version of this Use Case, only the following entityTypes are permitted for the Actor Monitored Unit:

- Compressor
- ElectricalImmersionHeater
- EVSE
- HeatPumpAppliance
- Inverter
- SmartEnergyAppliance
- SubMeterElectricity

This may change at a later version and be extended to further entityTypes.

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind one of the entityTypes listed in section 3.2.2.1.1, the Actor Monitored Unit SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
ElectricalConnection	1: M 2: M 3: M 4: M 5: M	electricalConnectionDescriptionListData	read (M). partial (R)
	1: M 2: M 3: M 4: M 5: M	electricalConnectionParameterDescriptionListData	read (M). partial (R)
Measurement	1: M 2: M 3: M 4: M 5: M	measurementDescriptionListData	read (M). partial (R)
	1: R 2: R 3: R 4: R 5: M	measurementConstraintsListData	read (M). partial (R)

	1: M 2: M 3: M 4: M 5: M	measurementListData	read (M). partial (R)
--	--------------------------------------	---------------------	--------------------------

Table 20: Feature Types and Functions used within this Use Case by the Actor Monitored Unit

For each of these Feature Types, the following rule applies: There SHALL be at maximum one Feature with the Feature Type in the Entity.

Note: As a consequence of the previous rule, an implementation may need to have Feature data from different Scenarios/Specializations or even Use Cases in a given Feature.

The Scenario number shows in which Scenarios the Monitored Unit acts as server and which Feature Types and Functions are relevant in each Scenario.

A detailed definition of the Elements and values that shall be supported in each Function is given in the following sub-sections.

Note: If in the table above "partial" read is not mentioned or is only optional, it still might be mandatory to support partial notifications. The details of "partial" support are described within the Scenario sections.

Note: The presence indications stated above are meant relative to the ones of the according Scenario stated in Table 1. I.e., if a Scenario is optional ("O") and a Feature Type is mandatory ("M"), the Feature Type need only be supported if the Scenario is supported, too.

Note: Further Features MAY be implemented on the same Entities; also, further Functions MAY be implemented in the used Entities.

3.2.2.2.2 Feature Type "ElectricalConnection"

3.2.2.2.2.1 Function "electricalConnectionDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M 2: M 3: M 4: M 5: M	ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData.		
1: M 2: M 3: M 4: M 5: M	electricalConnectionId	<ec1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	powerSupplyType	"ac"	

2: M			
3: M			
4: M			
5: M			
1: M	positiveEnergyDirection	"consume"	
2: M			
3: M			
4: M			
5: M			

Table 21: Content of Function "electricalConnectionDescriptionListData" at Actor Monitored Unit

3.2.2.2.2 Function "electricalConnectionParameterDescriptionListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	parameterId	<p1#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
1: M	measurementId	<m1#{1..1}->p1#1>	SHALL be set as FOREIGN IDENTIFIER.
1: M	voltageType	"ac"	
1: O	acMeasuredPhases	"abc" "ab" "bc" "ac" "a" "b" "c" (->p1#1)	If the Monitored Unit is connected to less than three phases, one of the other combinations like "a" or "ab" are allowed instead of "abc". The values "a", "b", and "c" are permitted if and only if only one phase is connected to the Monitored Unit.
1: O	acMeasuredInReferenceTo	"neutral"	
1: M	acMeasurementType	"real"	
1: O	acMeasurementVariant	"rms"	
1: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
1: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	parameterId	<p2#{1..3}->ec1#1>	SHALL be set as SUB IDENTIFIER.
1: M	measurementId	<m2#{1..3}->p2#{1..3}>	SHALL be set as FOREIGN IDENTIFIER.
1: M	voltageType	"ac"	
1: M	acMeasuredPhases	"a" (->p2#1)	[MPC-012/1]
		"b" (->p2#2)	[MPC-012/2]
		"c" (->p2#3)	[MPC-012/3]

1: M	acMeasuredInReferenceTo	"neutral"	
1: M	acMeasurementType	"real"	
1: M	acMeasurementVariant	"rms"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p3#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m3#{1..1}->p3#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: M	acMeasurementType	"real"	
2: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
2: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	parameterId	<p4#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
2: M	measurementId	<m4#{1..1}->p4#1>	SHALL be set as FOREIGN IDENTIFIER.
2: M	voltageType	"ac"	
2: M	acMeasurementType	"real"	
3: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
3: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	parameterId	<p5#{1..3}->ec1#1>	SHALL be set as SUB IDENTIFIER.
3: M	measurementId	<m5#{1..3}->p5#{1..3}>	SHALL be set as FOREIGN IDENTIFIER.
3: M	voltageType	"ac"	
3: M	acMeasuredPhases* ²	"a" (-><p5#1>)* ¹	[MPC-031/1]
		"b" (-><p5#2>)* ¹	[MPC-031/2]
		"c" (-><p5#3>)* ¹	[MPC-031/3]
3: M	acMeasurementType	"real"	[MPC-030/1]
3: M	acMeasurementVariant	"rms"	[MPC-030/2]
4: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
4: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	parameterId	<p6#{1..6}->ec1#1>	SHALL be set as SUB IDENTIFIER.
4: M	measurementId	<m6#{1..6}->p6#{1..6}>	SHALL be set as FOREIGN IDENTIFIER.
4: M	voltageType	"ac"	
4: M	acMeasuredPhases	"a" (-><p6#1>)	[MPC-041/1]
		"a" (-><p6#4>)	[MPC-041/4]
		"b" (-><p6#2>)	[MPC-041/2]
		"b" (-><p6#5>)	[MPC-041/5]
		"c" (-><p6#3>)	[MPC-041/3]
		"c" (-><p6#6>)	[MPC-041/6]
4: M	acMeasuredInReferenceTo	"a" (-><p6#6>)	[MPC-041/6]
		"b" (-><p6#4>)	[MPC-041/4]
		"c" (-><p6#5>)	[MPC-041/5]

		"neutral" (-><p6#1>)	[MPC-041/1]
		"neutral" (-><p6#2>)	[MPC-041/2]
		"neutral" (-><p6#3>)	[MPC-041/3]
4: M	acMeasurementType	"apparent"	
4: M	acMeasurementVariant	"rms"	
5: M	ElectricalConnection. electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData.		
5: M	electricalConnectionId	<ec1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	parameterId	<p7#{1..1}->ec1#1>	SHALL be set as SUB IDENTIFIER.
5: M	measurementId	<m7#{1..1}->p7#1>	SHALL be set as FOREIGN IDENTIFIER.
5: M	voltageType	"ac"	

Table 22: Content of Function "electricalConnectionParameterDescriptionListData" at Actor Monitored Unit

*¹: Each permitted value of the Element "acMeasuredPhases" SHALL NOT be used for more than one value of Element "parameterId".

*²: Only the connected phases SHALL be delivered. If only one phase is connected, only one of the values is provided by the Monitored Unit. This is not necessarily phase A, but could also be phase B or phase C.

3.2.2.2.3 Feature Type "Measurement"

3.2.2.2.3.1 Function "measurementDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	measurementType	"power"	
1: M	commodityType	"electricity"	
1: M	unit	"W"	
1: M	scopeType	"acPowerTotal"	
1: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
1: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	measurementType	"power"	
1: M	commodityType	"electricity"	
1: M	unit	"W"	
1: M	scopeType	"acPower"	
2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"acEnergyConsumed"	

2: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
2: M	measurementId	<m4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	measurementType	"energy"	
2: M	commodityType	"electricity"	
2: M	unit	"Wh"	
2: M	scopeType	"acEnergyProduced"	
3: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
3: M	measurementId	<m5#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	measurementType	"current"	
3: M	commodityType	"electricity"	
3: M	unit	"A"	
3: M	scopeType	"acCurrent"	
4: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
4: M	measurementId	<m6#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
4: M	measurementType	"voltage"	
4: M	commodityType	"electricity"	
4: M	unit	"V"	
4: M	scopeType	"acVoltage"	
5: M	Measurement. measurementDescriptionListData. measurementDescriptionData.		
5: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	measurementType	"frequency"	
5: M	commodityType	"electricity"	
5: M	unit	"Hz"	
5: M	scopeType	"acFrequency"	

Table 23: Content of Function "measurementDescriptionListData" at Actor Monitored Unit

3.2.2.2.3.2 Function "measurementConstraintsListData"

Scenario [...]: M/R/O [W]/[C]	Element	Value	[High-Level Mapping] Element and value rules
1: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMin. number		SHALL be used.
1: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
1: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMax. number		SHALL be used.
1: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.

1: R	valueStepSize.		SHOULD be used. [Scaled number rules]
1: M	valueStepSize. number		SHALL be used.
1: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
1: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
1: M	measurementId	<m2#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
1: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMin. number		SHALL be used.
1: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
1: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
1: M	valueRangeMax. number		SHALL be used.
1: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
1: R	valueStepSize.		SHOULD be used. [Scaled number rules]
1: M	valueStepSize. number		SHALL be used.
1: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

2: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMin. number		SHALL be used.
2: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
2: M	valueRangeMax. number		SHALL be used.
2: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
2: R	valueStepSize.		SHOULD be used. [Scaled number rules]
2: M	valueStepSize. number		SHALL be used.
2: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
3: M	measurementId	<m5#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
3: R	valueRangeMin.		[MPC-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMin. number		SHALL be used.
3: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueRangeMax.		[MPC-001] SHOULD be used. [Scaled number rules]
3: M	valueRangeMax. number		SHALL be used.
3: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
3: R	valueStepSize.		SHOULD be used. [Scaled number rules]
3: M	valueStepSize. number		SHALL be used.
3: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
4: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
4: M	measurementId	<m6#{1..6}>	SHALL be set as PRIMARY IDENTIFIER.
4: R	valueRangeMin.		[MPC-002] SHOULD be used. [Scaled number rules]
4: M	valueRangeMin. number		SHALL be used.
4: O	valueRangeMin. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueRangeMax.		[MPC-002]

			SHOULD be used. [Scaled number rules]
4: M	valueRangeMax. number		SHALL be used.
4: O	valueRangeMax. scale		MAY be used. If absent, a default value of "0" applies.
4: R	valueStepSize.		SHOULD be used. [Scaled number rules]
4: M	valueStepSize. number		SHALL be used.
4: O	valueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
5: R	Measurement. measurementConstraintsListData. measurementConstraintsData.		
5: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: R	valueRangeMin.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMin. number		SHALL be used.
5: O	valueRangeMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueRangeMax.		SHOULD be used. [Scaled number rules]
5: M	valueRangeMax. number		SHALL be used.
5: O	valueRangeMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: R	valueStepSize.		SHOULD be used. [Scaled number rules]
5: M	valueStepSize. number		SHALL be used.
5: O	valueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.

Table 24: Content of Function "measurementConstraintsListData" at Actor Monitored Unit

3.2.2.2.3.3 Function "measurementListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	Measurement. measurementListData. measurementData.		
1: M	measurementId	<m1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
1: O	timestamp	<t#{1..1}>->m1#1>	MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
1: M	value.		[MPC-001], [MPC-011]

			[Measurement value rules] [Scaled number rules]
1: M	value. number		SHALL be used.
1: O	value. scale		MAY be used. If absent, a default value of "0" applies.
1: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: R	valueState		[MPC-003] [Value state rules]
1: M	Measurement. measurementListData. measurementData.		
1: M	measurementId	<m2#{1..3}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
1: O	timestamp	<t#{1..1}->m2#{1..3}>	MAY be used. Within this Use Case, only the newest measurement value SHALL be stated. Additional historical values are forbidden.
1: M	value.		[MPC-001], [MPC-012] See [Measurement value rules]. See [Scaled number rules].
1: M	value. number		SHALL be used.
1: O	value. scale		MAY be used. If absent, a default value of "0" applies.
1: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
1: R	valueState		[MPC-003] [Value state rules]
2: M	Measurement. measurementListData. measurementData.		
2: M	measurementId	<m3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#{1..1}->m3#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MPC-001], [MPC-021] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: O	evaluationPeriod.		[Evaluation period rules]
2: M	evaluationPeriod. startTime		
2: M	evaluationPeriod. endTime		
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MPC-003] [Value state rules]
2: M	Measurement. measurementListData. measurementData.		

2: M	measurementId	<m4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
2: O	timestamp	<t#(1..1)->m4#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
2: M	value.		[MPC-001], [MPC-022] [Measurement value rules] [Scaled number rules]
2: M	value. number		SHALL be used.
2: O	value. scale		MAY be used. If absent, a default value of "0" applies.
2: O	evaluationPeriod.		[Evaluation period rules]
2: M	evaluationPeriod. startTime		
2: M	evaluationPeriod. endTime		
2: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
2: R	valueState		[MPC-003] [Value state rules]
3: M	Measurement. measurementListData. measurementData.		
3: M	measurementId	<m5#(1..3)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
3: O	timestamp	<t#(1..1)->m5#(1..3)>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
3: M	value.		[MPC-001], [MPC-031] [Measurement value rules] [Scaled number rules]
3: M	value. number		SHALL be used.
3: O	value. scale		MAY be used. If absent, a default value of "0" applies.
3: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
3: R	valueState		[MPC-003] [Value state rules]
4: M	Measurement. measurementListData. measurementData.		
4: M	measurementId	<m6#(1..6)>	SHALL be set as PRIMARY IDENTIFIER.
4: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
4: O	timestamp	<t#(1..1)->m6#(1..6)>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
4: M	value.		[MPC-002], [MPC-041] [Measurement value rules] [Scaled number rules]
4: M	value. number		SHALL be used.
4: O	value. scale		MAY be used. If absent, a default value of "0" applies.
4: M	valueSource	"measuredValue"	

		"calculatedValue"	
		"empiricalValue"	
4: R	valueState		[MPC-003] [Value state rules]
5: M	Measurement. measurementListData. measurementData.		
5: M	measurementId	<m7#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
5: M	valueType	"value"	SHALL be set as SUB IDENTIFIER.
5: O	timestamp	<t#{1..1}->m7#1>	MAY be used. Only the newest measurement value SHALL be stated. Additional historical values are forbidden.
5: M	value.		[MPC-051] [Measurement value rules] [Scaled number rules]
5: M	value. number		SHALL be used.
5: O	value. scale		SHALL be interpreted. If absent, a default value of "0" applies.
5: M	valueSource	"measuredValue"	
		"calculatedValue"	
		"empiricalValue"	
5: R	valueState		[MPC-003] [Value state rules]

Table 25: Content of Function "measurementListData" at Actor Monitored Unit

3.2.2.3 Client data - Specializations

As this Actor has only server functionality, no Specializations are described within this section.

3.3 Pre-Scenario communication

3.3.1 General information

The Pre-Scenario communication is needed if a client does not know the corresponding addresses on the server or if the required subscriptions or bindings are not active. In this case certain general communication mechanisms SHALL be used within SPINE:

- Detailed discovery: allows to discover resource addresses.
- Binding: allows to bind to resource address, which is frequently necessary to obtain write privileges.
- Subscription: allows to subscribe to resource addresses, which is necessary to receive unsolicited notifications if a resource changes during runtime.

It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription needs to occur.
- E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs to occur only once for all Use Cases.

Depending on which Entity, Feature and Functions are used within a Scenario the payload of the corresponding messages may slightly differ, but the basic principles and messages used stay the same.

The subsequent messages SHALL be exchanged for those parts that have not already been performed since the current connection is established or if those parts are outdated for another reason (e.g. if the detailed discovery is needed, but the bindings and subscriptions are still active from a previous connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario communication parts are up to date, this section MAY be skipped, and the implementation can proceed as described in the corresponding "Scenario communication" sections.

After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-Scenario communication SHALL be performed as well. There may be circumstances where messages from the Pre-Scenario communication may be exchanged again.

Often the necessary messages of different Scenarios can be combined, so that only one single message is needed instead of multiple messages for the different Scenarios. This also is the case for the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery is necessary, as well as only one subscription request or binding request is needed for each Feature. Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or binding is necessary.

3.3.2 Detailed discovery

For the functionality where a client already has current detailed discovery information (i.e. independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

Otherwise, the following procedure SHALL be performed in the given order:

1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL acquire a subscription according to the figure provided within this sub-section.
2. A client SHALL read the detailed discovery information according to the figure provided within this sub-section. It SHALL keep the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case that are needed by the client. This means that a client may choose how to store the necessary information. E.g. a client Actor can store the information how to address the necessary Features of the implemented Scenarios but may discard the Entity information.
3. If and as long as a client has a subscription to the detailed discovery information of an Actor and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed discovery information) the received information as far as it concerns mandatory and supported optional Entity Types, Feature Types and Functions of this Use Case.

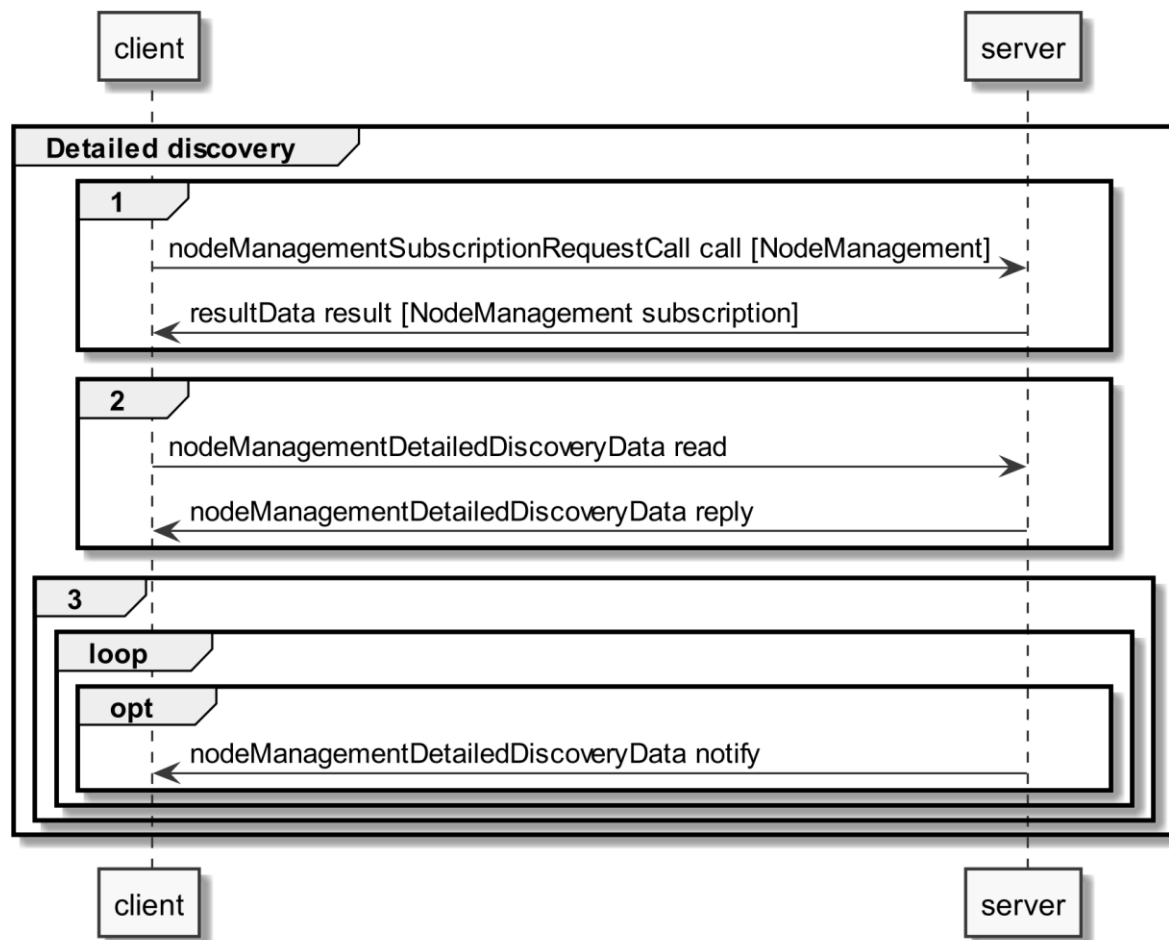


Figure 6: Pre-Scenario communication - Detailed discovery sequence diagram

If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message was received successfully.

If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply" message contains at least the mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as the used Functions and their "possible operations" described in section 3.2 and its sub-sections.

If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...] overview" diagrams as well as all needed Functions and their "possible operations" described in section 3.2 and its sub-sections. However, as soon as the functionality is available it will be announced via a "nodeManagementDetailedDiscoveryData notify" message.

For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial read with separated Selectors that may use one of the following Elements:

- entityType
- featureType

Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case may be discovered. However, only Features and Entities that fulfil the hierarchical order as described within the Actors' sections shall be considered for this Use Case.

A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify SHOULD also be supported with separated Selectors that may use one of the following Elements:

- entityAddress
- featureAddress

3.3.3 Binding

If binding is required by a Scenario that uses Features with writeable or changeable data, the server SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs, the client SHALL create a binding to the corresponding Feature. For this the nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:

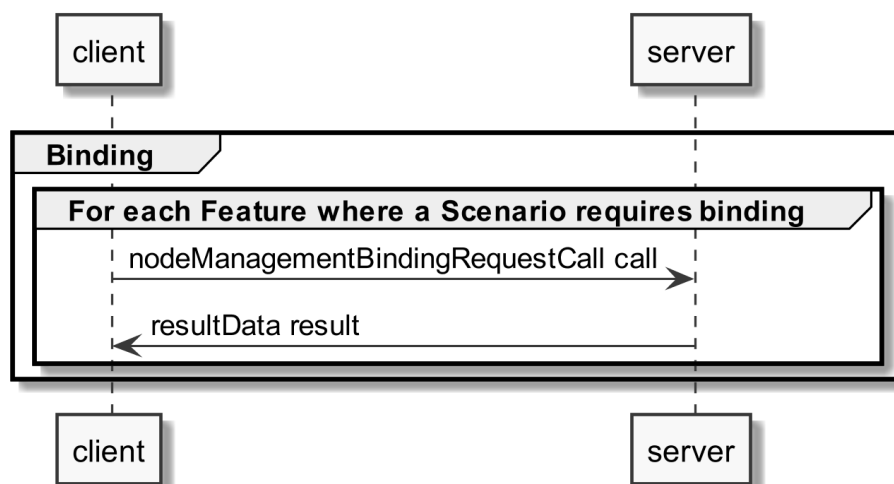


Figure 7: Pre-Scenario communication - Binding sequence diagram

If functionality is added or removed dynamically, binding may not be possible at all times on the required Functions. A client SHALL retry to create a binding again when receiving according updated detailed discovery information.

3.3.4 Subscription

A server SHALL support subscription for all Features that contain readable data that may change during runtime. The client SHALL create a subscription for all Features that the client wants to read. For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following sequence diagram:

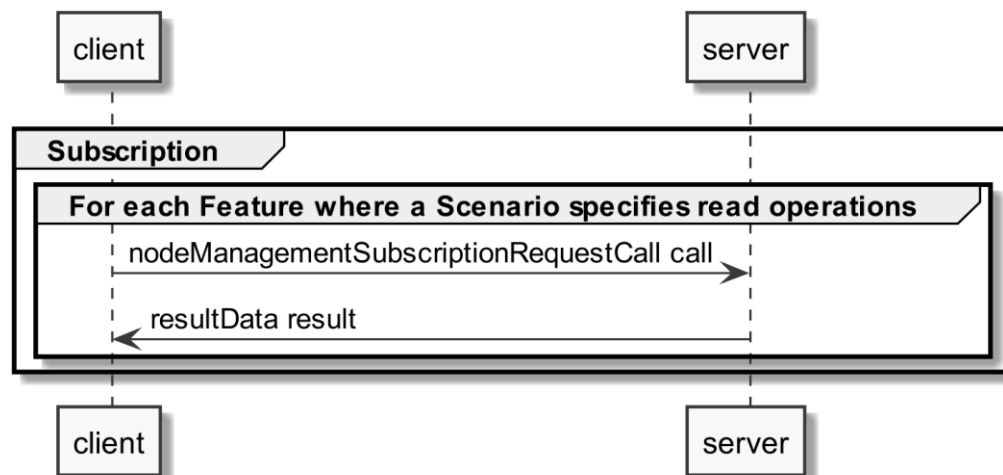


Figure 8: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client **SHOULD** read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client **SHALL** retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features **SHALL** be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Monitor power

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Binding **SHOULD NOT** be used for this Scenario.
3. **Subscription:** Actors **SHALL** create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication **SHALL** start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as

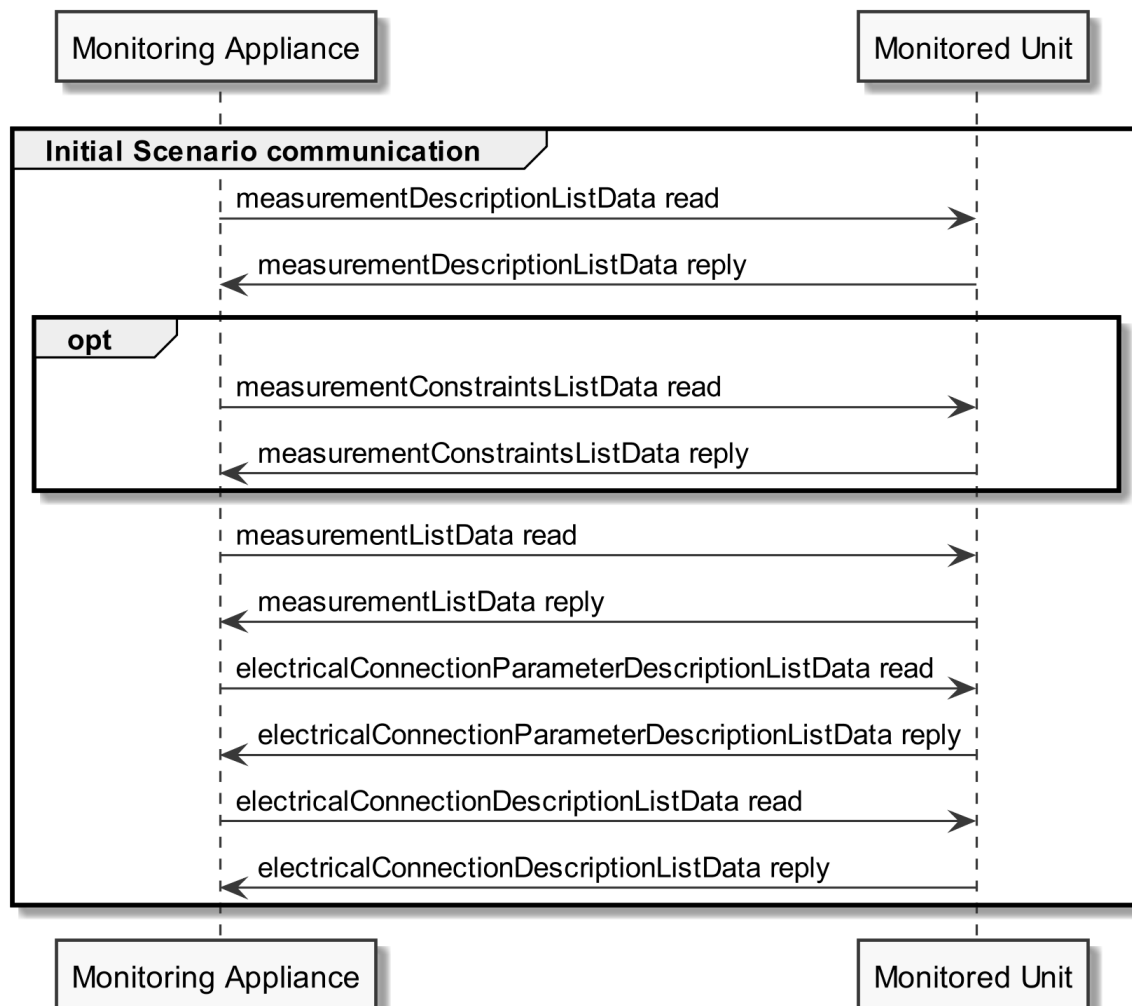
1074 soon as the address of a required resource is known, the Initial Scenario communication for this
 1075 resource MAY start already, even if the addresses of other required resources are not known yet.

1076 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
 1077 communication is triggered again for those resources.

1078

1079 **3.4.1.2 Initial Scenario communication**

1080 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1081 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1082 resources may have changed in the meantime:



1083

1084 *Figure 9: Scenario 1 - Initial Scenario communication sequence diagram*

1085 **Partial data information regarding [MPC-011]:**

1086 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
 1087 Selector:

- 1088 - scopeType = "acPowerTotal"

1089 The measurementConstraintsListData read, measurementListData read and
 1090 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
 1091 the following Selector:

1092 - measurementId (derived from the measurementDescriptionListData reply)

1093 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
 1094 following Selector:

1095 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
 1096 reply)

1097 Note: If partial read is not supported, a full read SHALL be performed.

1098

1099 **Partial data information regarding [MPC-012]:**

1100 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
 1101 Selector:

1102 - scopeType = "acPower"

1103 The measurementConstraintsListData read, measurementListData read and
 1104 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
 1105 the following Selector:

1106 - measurementId (derived from the measurementDescriptionListData reply)

1107 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
 1108 following Selector:

1109 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
 1110 reply)

1111 Note: If partial read is not supported, a full read SHALL be performed.

1112

1113 The following table shows where the required content of the messages from the sequence diagram is
 1114 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 23	1
measurementConstraintsListData reply	Table 24	1
measurementListData reply	Table 25	1
electricalConnectionParameterDescriptionListData reply	Table 22	1
electricalConnectionDescriptionListData reply	Table 21	1

1115 *Table 26: Initial Scenario communication content references for Scenario 1*

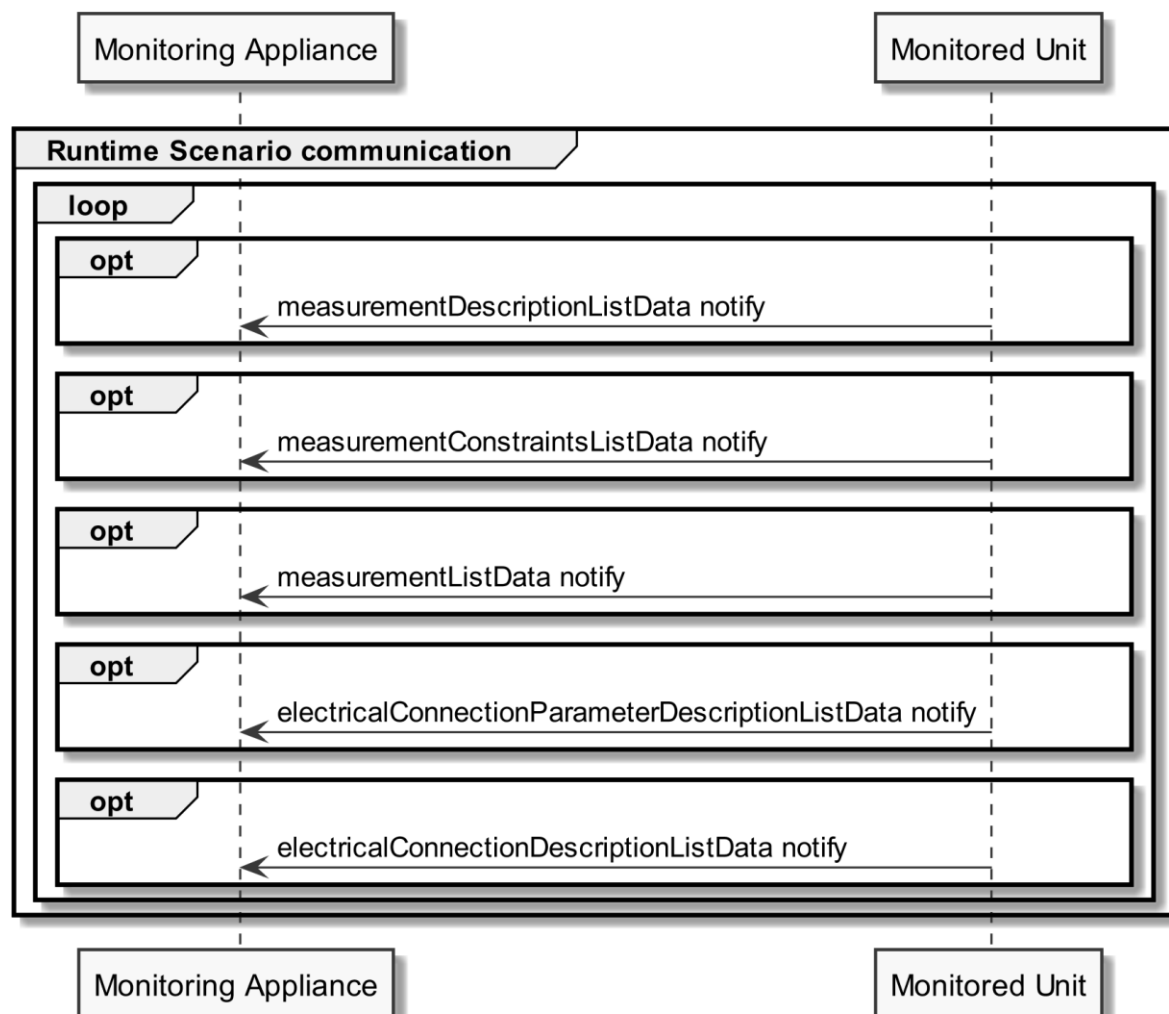
1116 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1117 provided completely, but later during Runtime Scenario communication.

1118

1119 **3.4.1.3 Runtime Scenario communication**

1120 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1121 during runtime.

1122 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1123 in the following figure:



1124

1125 *Figure 10: Scenario 1 - Runtime Scenario communication sequence diagram*

1126 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1127 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1128 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1129 Scenario.

1130 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1131 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

1132 - measurementId

1133 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
1134 be supported with the Selectors:

- 1135 - electricalConnectionId
- 1136 - parameterId
- 1137 - measurementId

1138 Note: To interpret partial notification messages correctly the information obtained during the Initial
1139 Scenario communication phase is required.

1140 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
1141 not be evaluated.

1142

1143 The following table shows where the required content of the messages of the sequence diagram is
1144 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 23	1
measurementConstraintsListData notify	Table 24	1
measurementListData notify	Table 25	1
electricalConnectionParameterDescriptionListData notify	Table 22	1
electricalConnectionDescriptionListData notify	Table 21	1

1145 *Table 27: Runtime Scenario communication content references for Scenario 1*

1146

1147 **3.4.1.4 Additional information**

1148 The measured power always refers to the whole Entity and all sub-Entities behind it.

1149

1150 **3.4.2 Scenario 2 - Monitor energy**

1151 **3.4.2.1 Pre-Scenario communication**

- 1152 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
1153 of the server Features used in the Initial Scenario communication. If the address of a
1154 particular server Feature is not known, the detailed discovery must be used, as described in
1155 section 3.3.2.
- 1156 2. **Binding:** Binding SHOULD NOT be used for this Scenario.
- 1157 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
1158 the corresponding Actor within this Scenario, as described in section 3.3.4.

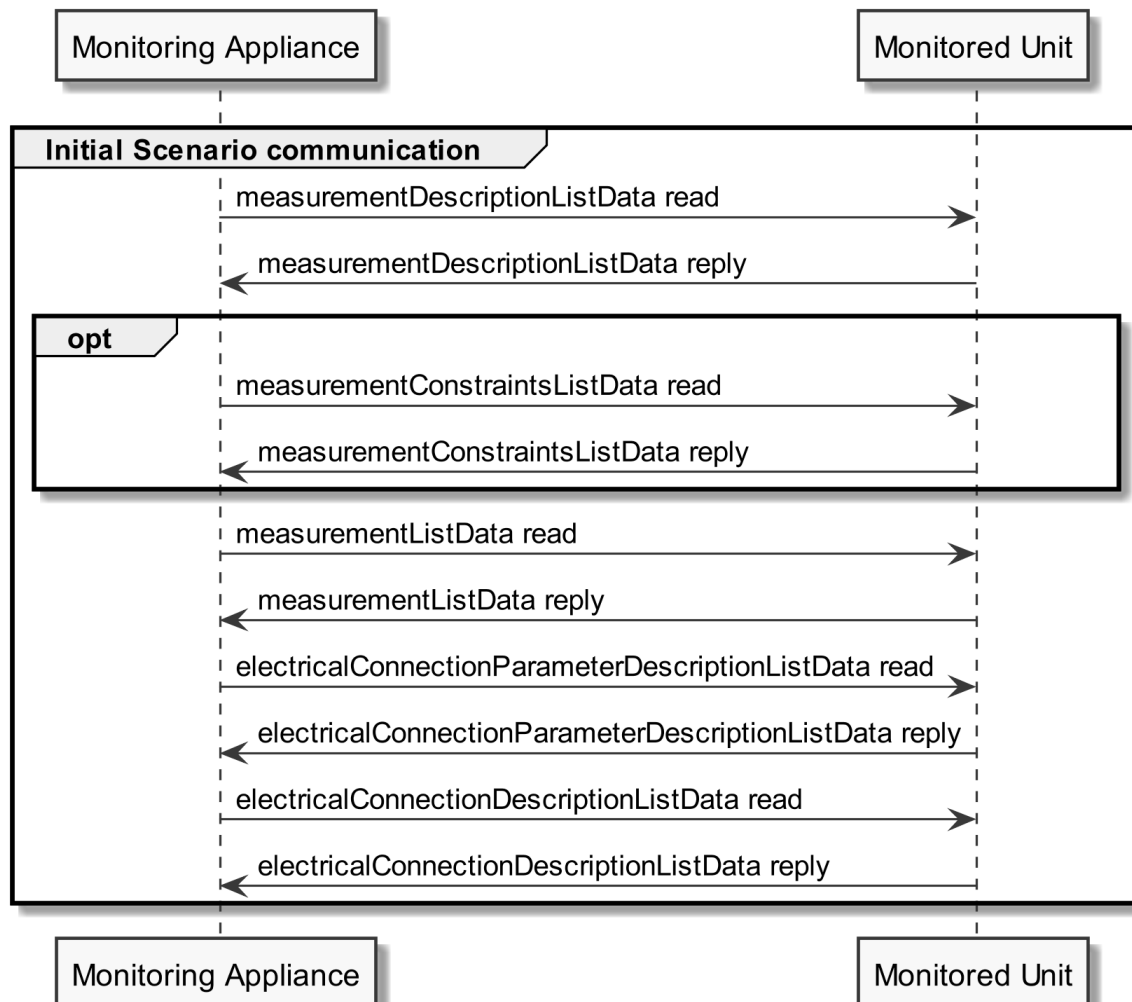
1159 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
1160 are known and the necessary binding and subscription procedures have been finished. However, as
1161 soon as the address of a required resource is known, the Initial Scenario communication for this
1162 resource MAY start already, even if the addresses of other required resources are not known yet.

1163 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1164 communication is triggered again for those resources.

1165

1166 **3.4.2.2 Initial Scenario communication**

1167 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
 1168 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
 1169 resources may have changed in the meantime:



1170

1171 *Figure 11: Scenario 2 - Initial Scenario communication sequence diagram*1172 **Partial data information regarding [MPC-021]:**

1173 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
 1174 Selector:

- 1175 - scopeType = "acEnergyConsumed"

1176 The measurementConstraintsListData read, measurementListData read and
 1177 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
 1178 the following Selector:

- 1179 - measurementId (derived from the measurementDescriptionListData reply)

1180 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1181 following Selector:

- 1182 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1183 reply)

1184 Note: If partial read is not supported, a full read SHALL be performed.

1185

1186 **Partial data information regarding [MPC-022]:**

1187 The measurementDescriptionListData read SHOULD be a "partial" read operation with the following
1188 Selector:

- 1189 - scopeType = "acEnergyProduced"

1190 The measurementConstraintsListData read, measurementListData read and
1191 electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with
1192 the following Selector:

- 1193 - measurementId (derived from the measurementDescriptionListData reply)

1194 The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the
1195 following Selector:

- 1196 - electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData
1197 reply)

1198 Note: If partial read is not supported, a full read SHALL be performed.

1199

1200 The following table shows where the required content of the messages from the sequence diagram is
1201 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 23	2
measurementConstraintsListData reply	Table 24	2
measurementListData reply	Table 25	2
electricalConnectionParameterDescriptionListData reply	Table 22	2
electricalConnectionDescriptionListData reply	Table 21	2

1202 *Table 28: Initial Scenario communication content references for Scenario 2*

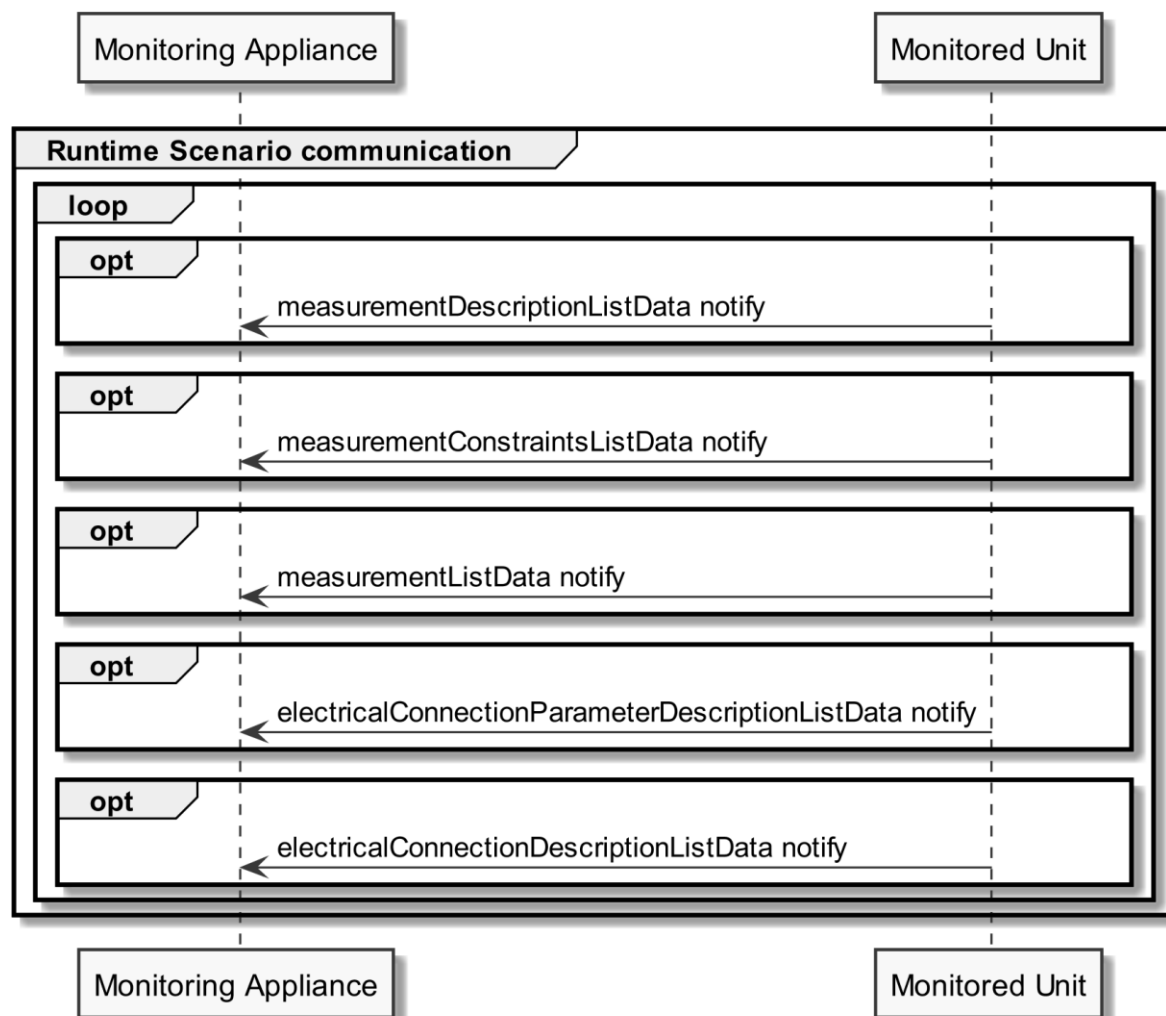
1203 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1204 provided completely, but later during Runtime Scenario communication.

1205

1206 **3.4.2.3 Runtime Scenario communication**

1207 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1208 during runtime.

1209 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1210 in the following figure:



1211

1212 *Figure 12: Scenario 2 - Runtime Scenario communication sequence diagram*

1213 Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime.
 1214 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1215 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1216 Scenario.

1217 For measurementDescriptionListData notify, measurementConstraintsListData notify and
 1218 measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- 1219 - measurementId

1220 For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD
 1221 be supported with the Selectors:

- 1222 - electricalConnectionId
 1223 - parameterId
 1224 - measurementId

1225 Note: To interpret partial notification messages correctly the information obtained during the Initial
1226 Scenario communication phase is required.

1227 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
1228 not be evaluated.

1229

1230 The following table shows where the required content of the messages of the sequence diagram is
1231 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 23	2
measurementConstraintsListData notify	Table 24	2
measurementListData notify	Table 25	2
electricalConnectionParameterDescriptionListData notify	Table 22	2
electricalConnectionDescriptionListData notify	Table 21	2

1232 *Table 29: Runtime Scenario communication content references for Scenario 2*

1233

1234 **3.4.2.4 Additional information**

1235 The measured energy always refers to the whole Entity and all sub-Entities behind it.

1236

1237 **3.4.3 Scenario 3 - Monitor current**

1238 **3.4.3.1 Pre-Scenario communication**

- 1239 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
1240 of the server Features used in the Initial Scenario communication. If the address of a
1241 particular server Feature is not known, the detailed discovery must be used, as described in
1242 section 3.3.2.
- 1243 2. **Binding:** Binding SHOULD NOT be used for this Scenario.
- 1244 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
1245 the corresponding Actor within this Scenario, as described in section 3.3.4.

1246 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
1247 are known and the necessary binding and subscription procedures have been finished. However, as
1248 soon as the address of a required resource is known, the Initial Scenario communication for this
1249 resource MAY start already, even if the addresses of other required resources are not known yet.

1250 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1251 communication is triggered again for those resources.

1252

1253 **3.4.3.2 Initial Scenario communication**

1254 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
1255 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
1256 resources may have changed in the meantime:

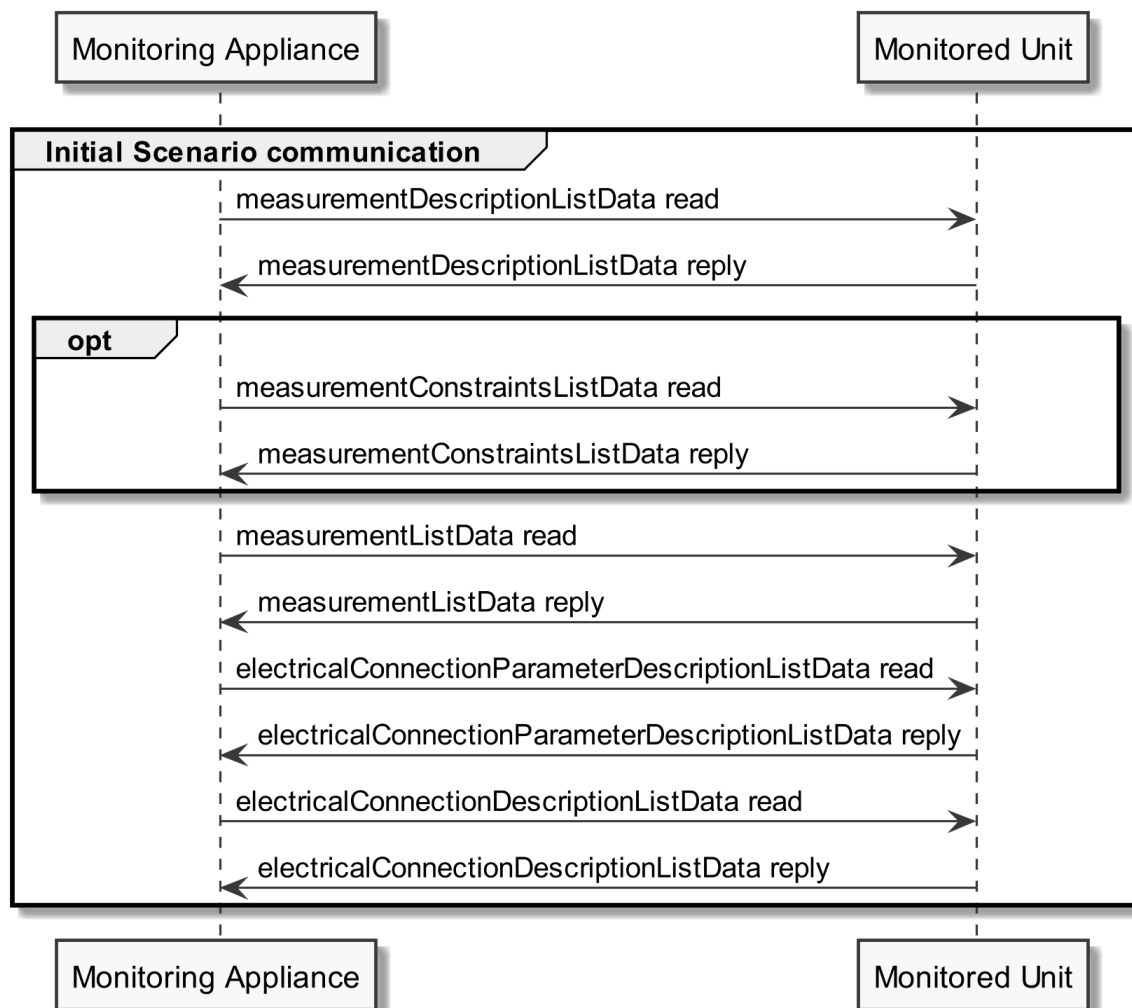


Figure 13: Scenario 3 - Initial Scenario communication sequence diagram

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "acCurrent"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

1272 The following table shows where the required content of the messages from the sequence diagram is
 1273 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 23	3
measurementConstraintsListData reply	Table 24	3
measurementListData reply	Table 25	3
electricalConnectionParameterDescriptionListData reply	Table 22	3
electricalConnectionDescriptionListData reply	Table 21	3

1274 *Table 30: Initial Scenario communication content references for Scenario 3*

1275 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1276 provided completely, but later during Runtime Scenario communication.

1277

1278 **3.4.3.3 Runtime Scenario communication**

1279 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1280 during runtime.

1281 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1282 in the following figure:

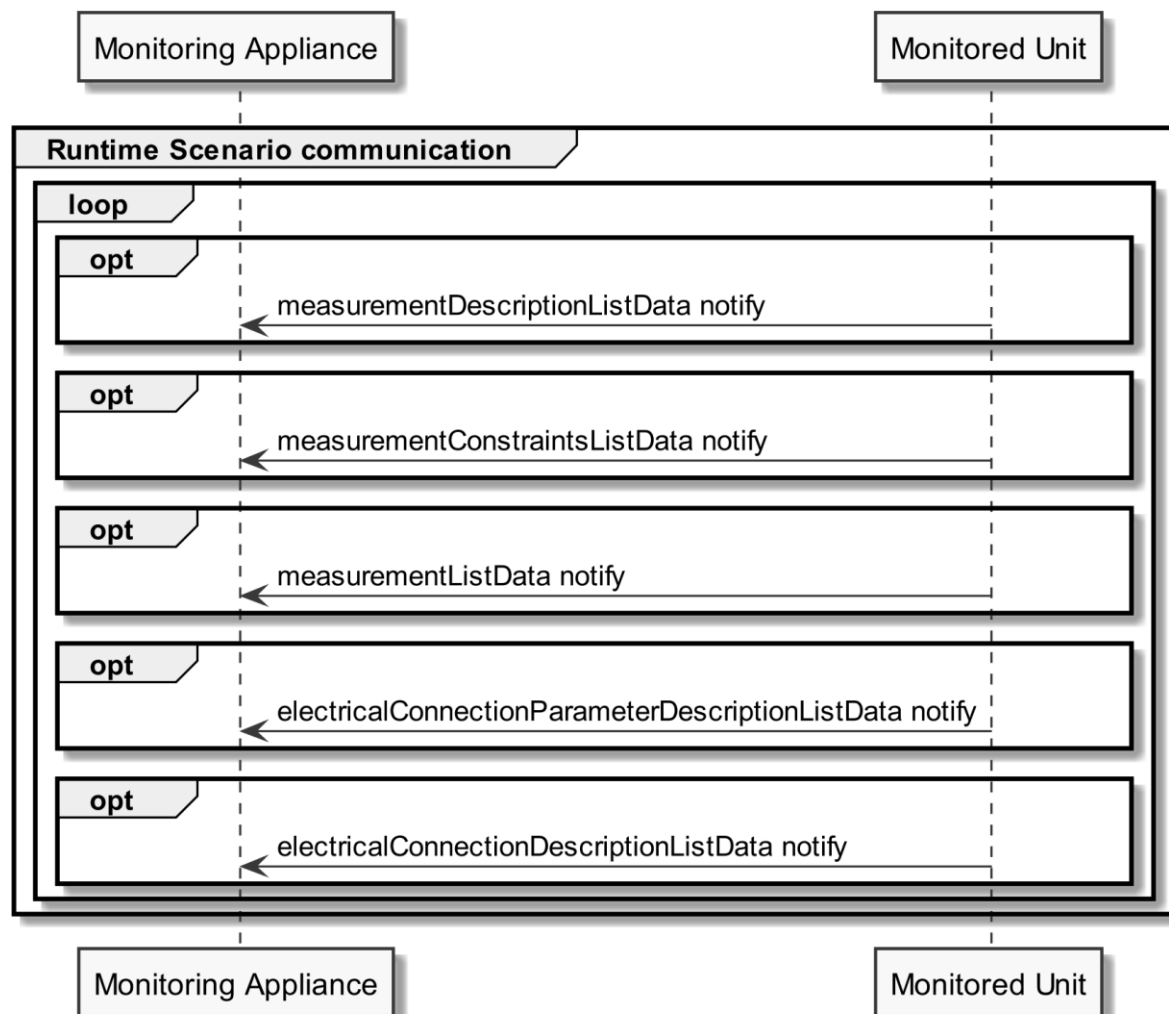


Figure 14: Scenario 3 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For measurementDescriptionListData notify, measurementConstraintsListData notify and measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- measurementId

For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- measurementId

Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

1299 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
1300 not be evaluated.

1301

1302 The following table shows where the required content of the messages of the sequence diagram is
1303 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 23	3
measurementConstraintsListData notify	Table 24	3
measurementListData notify	Table 25	3
electricalConnectionParameterDescriptionListData notify	Table 22	3
electricalConnectionDescriptionListData notify	Table 21	3

1304 *Table 31: Runtime Scenario communication content references for Scenario 3*

1305

1306 **3.4.3.4 Additional information**

1307 The measured current always refers to the whole Entity and all sub-Entities behind it.

1308

1309 **3.4.4 Scenario 4 - Monitor voltage**

1310 **3.4.4.1 Pre-Scenario communication**

- 1311 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
1312 of the server Features used in the Initial Scenario communication. If the address of a
1313 particular server Feature is not known, the detailed discovery must be used, as described in
1314 section 3.3.2.
- 1315 2. **Binding:** Binding SHOULD NOT be used for this Scenario.
- 1316 3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for
1317 the corresponding Actor within this Scenario, as described in section 3.3.4.

1318 The Initial Scenario communication SHALL start at the latest when the required resources on an Actor
1319 are known and the necessary binding and subscription procedures have been finished. However, as
1320 soon as the address of a required resource is known, the Initial Scenario communication for this
1321 resource MAY start already, even if the addresses of other required resources are not known yet.

1322 If required resources are removed and added again, they are re-discovered, and the Initial Scenario
1323 communication is triggered again for those resources.

1324

1325 **3.4.4.2 Initial Scenario communication**

1326 Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped,
1327 the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding
1328 resources may have changed in the meantime:

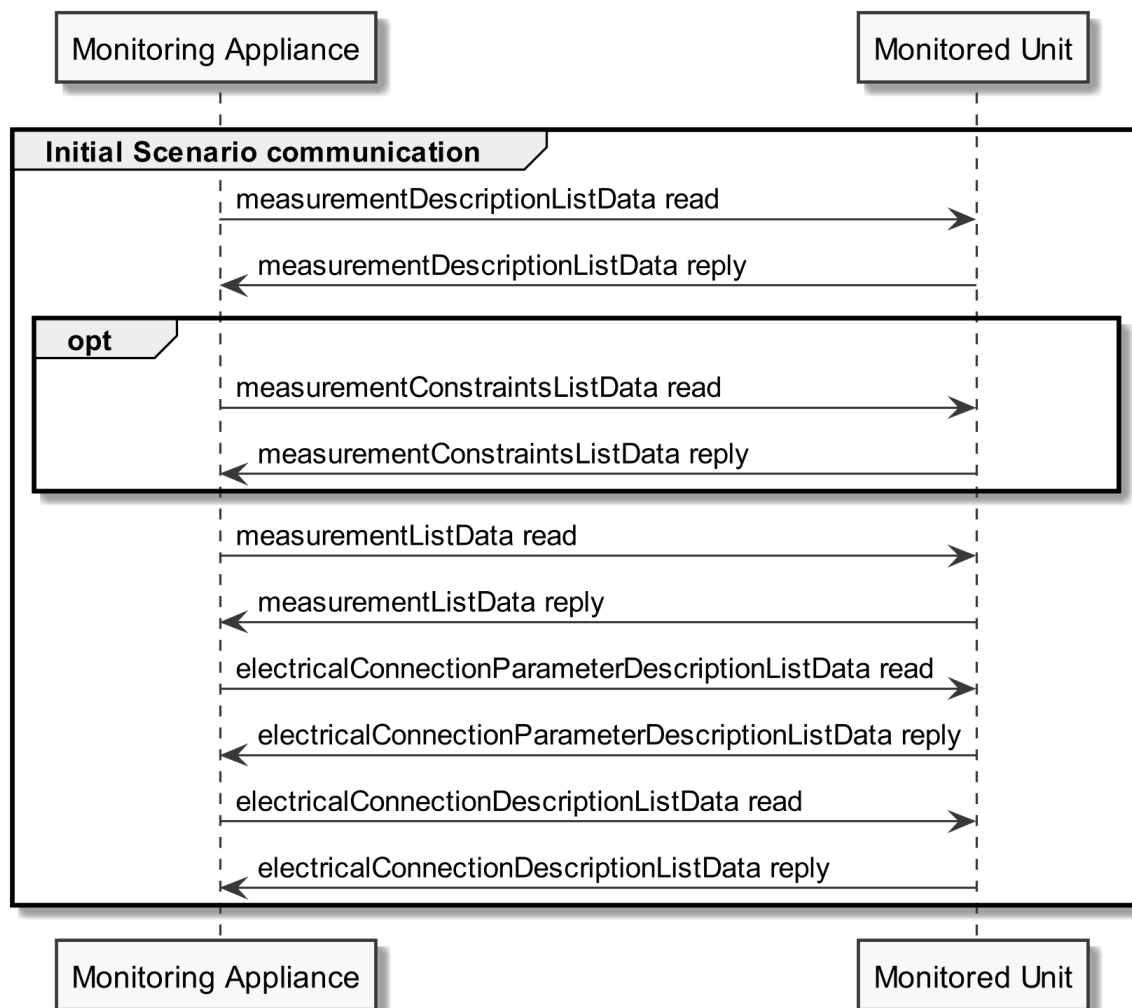


Figure 15: Scenario 4 - Initial Scenario communication sequence diagram

Partial data information regarding [MPC-041]:

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "acVoltage"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

1345

1346 The following table shows where the required content of the messages from the sequence diagram is
1347 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 23	4
measurementConstraintsListData reply	Table 24	4
measurementListData reply	Table 25	4
electricalConnectionParameterDescriptionListData reply	Table 22	4
electricalConnectionDescriptionListData reply	Table 21	4

1348 *Table 32: Initial Scenario communication content references for Scenario 4*

1349 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1350 provided completely, but later during Runtime Scenario communication.

1351

1352 **3.4.4.3 Runtime Scenario communication**

1353 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1354 during runtime.

1355 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
1356 in the following figure:

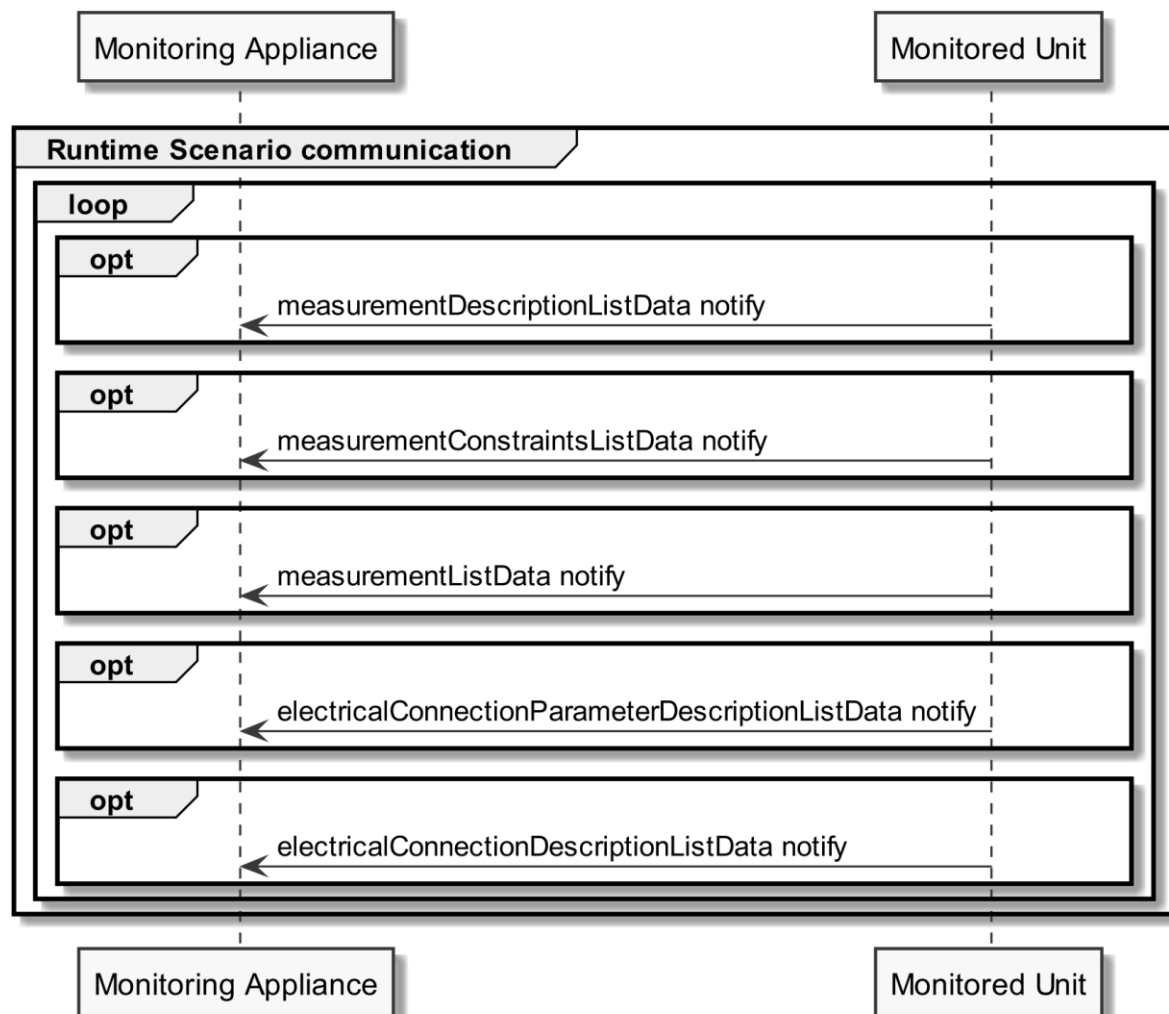


Figure 16: Scenario 4 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For measurementDescriptionListData notify, measurementConstraintsListData notify and measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- measurementId

For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- measurementId

Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 23	4
measurementConstraintsListData notify	Table 24	4
measurementListData notify	Table 25	4
electricalConnectionParameterDescriptionListData notify	Table 22	4
electricalConnectionDescriptionListData notify	Table 21	4

Table 33: Runtime Scenario communication content references for Scenario 4

3.4.4.4 Additional information

The measured voltage always refers to the whole Entity and all sub-Entities behind it.

3.4.5 Scenario 5 - Monitor frequency

3.4.5.1 Pre-Scenario communication

- Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
- Binding:** Binding SHOULD NOT be used for this Scenario.
- Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.5.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

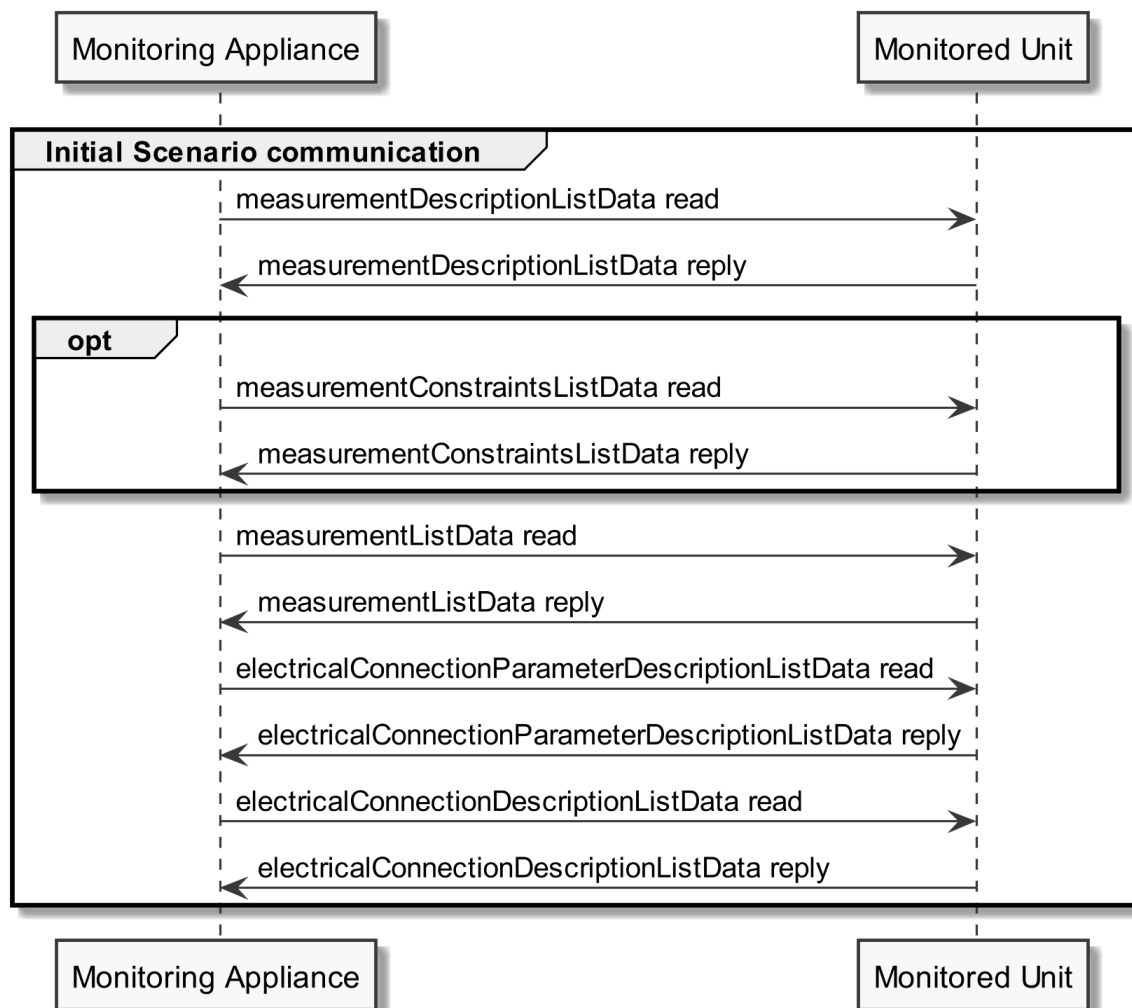


Figure 17: Scenario 5 - Initial Scenario communication sequence diagram

The measurementDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- scopeType = "acFrequency"

The measurementConstraintsListData read, measurementListData read and electricalConnectionParameterDescriptionListData read SHOULD be "partial" read operations with the following Selector:

- measurementId (derived from the measurementDescriptionListData reply)

The electricalConnectionDescriptionListData read SHOULD be a "partial" read operation with the following Selector:

- electricalConnectionId (derived from the electricalConnectionParameterDescriptionListData reply)

Note: If partial read is not supported, a full read SHALL be performed.

1418 The following table shows where the required content of the messages from the sequence diagram is
 1419 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData reply	Table 23	5
measurementConstraintsListData reply	Table 24	5
measurementListData reply	Table 25	5
electricalConnectionParameterDescriptionListData reply	Table 22	5
electricalConnectionDescriptionListData reply	Table 21	5

1420 *Table 34: Initial Scenario communication content references for Scenario 5*

1421 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1422 provided completely, but later during Runtime Scenario communication.

1423

1424 **3.4.5.3 Runtime Scenario communication**

1425 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1426 during runtime.

1427 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1428 in the following figure:

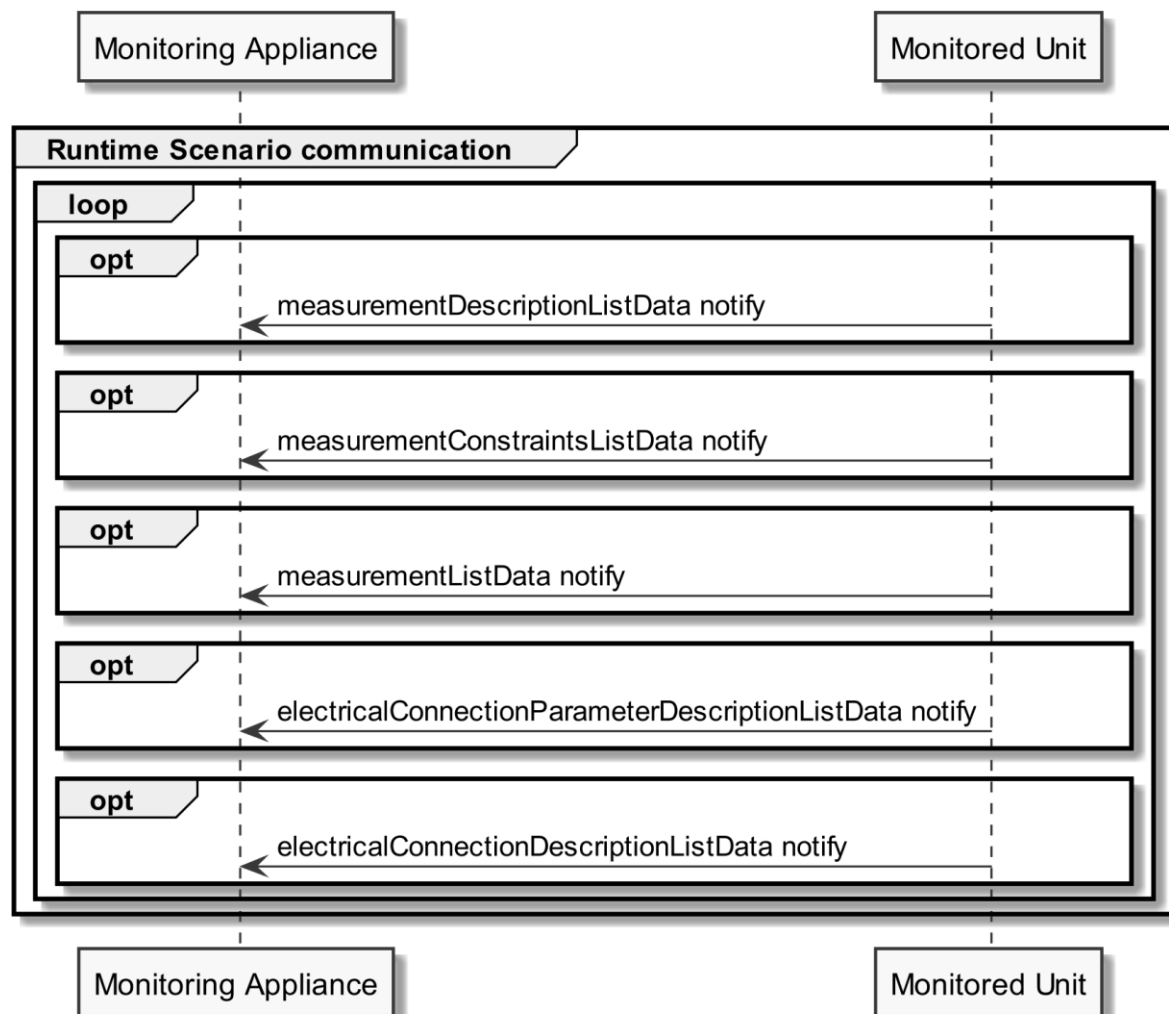


Figure 18: Scenario 5 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "measurementListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For measurementDescriptionListData notify, measurementConstraintsListData notify and measurementListData notify "partial" delete notifications SHOULD be supported with the Selector:

- measurementId

For electricalConnectionParameterDescriptionListData notify "partial" delete notifications SHOULD be supported with the Selectors:

- electricalConnectionId
- parameterId
- measurementId

Note: To interpret partial notification messages correctly the information obtained during the Initial Scenario communication phase is required.

1445 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
1446 not be evaluated.

1447

1448 The following table shows where the required content of the messages of the sequence diagram is
1449 described:

Message name from sequence diagram	Content description in table	Scenario number in table
measurementDescriptionListData notify	Table 23	5
measurementConstraintsListData notify	Table 24	5
measurementListData notify	Table 25	5
electricalConnectionParameterDescriptionListData notify	Table 22	5
electricalConnectionDescriptionListData notify	Table 21	5

1450 *Table 35: Runtime Scenario communication content references for Scenario 5*

1451

1452 **3.4.5.4 Additional information**

1453 The measured voltage always refers to the whole Entity and all sub-Entities behind it.

1454